nMDS and PERMANOVA

ENVX2001 Applied Statistical Methods

Semester 1, 2026

Readings

Quinn, G. P. and Keough, M. J. (2002, 2023) *Experimental Design and Data Analysis for Biologists*. Cambridge University Press (1st and 2nd edition).

Han, S. Y., Filippi, P., Román Dobarco, M., Harianto, J., Crowther M. S., and Bishop, T. F. A. (2023). Multivariate analysis for soil science. In *'Encyclopedia of Soils in the Environment (Second Edition)'*. (Ed. M. J. Goss and M. Oliver) pp. 499-508. Academic Press: Oxford.

Learning outcomes

At the end of this practical students should be able to:

1. Calculate a similarity matrix or transformed data
2. Plot an ordination using Non-metric Multidimensional Scaling (nMDS)
3. Statistically test differences between factors using a ANOSIM and PERMANOVA
4. Determine what contributes most to the differences using a SIMPER analysis

Working directory Set the working directory for this tutorial and ensure you copy all data into this directory and save your code into the directory. setwd("C:/Users/.")

Exercise 1: Copepod Communities in Norway

It is thought that copepod (a marine invertebrate) assemblages in Solbergstrand, Norway are affected by nutrient levels. There were 4 spatially independent sites for each treatment in;

a) Control sites (C),

b) Low nutrient sites(L) and

c) High nutrient sites (H).

We want to know if there is a pattern in species assemblages using nMDS, whether there is a statistically significant difference between treatments using ANOSIM and what species contribute to these differences via SIMPER.

First you need to load the packages vegan and MASS and file CopepodData.csv

CODE
```
library(vegan)
```

OUTPUT
```
Loading required package: permute
```

CODE
```
library(MASS)

CopepodData <- read.csv("data/CopepodData.csv", header = TRUE)
```

Now the data is entered, let's apply a 4th root transformation. This will take the emphasis from the more common species. Other transformations include the square root transformation and the log transformation. Other transformations included square root, presence/absence and log. You can also standardise the data if there are differences in the amount of sampling (which there isn't here).

CODE
```
TransCopepodData <- (CopepodData[,2:20])^(1/4)
```

Now let's generate a Bray-Curtis dissimilarity matrix on this transformed data and perform a nMDS. Bray-Curtis similarities are useful for this kind of data, as they did not group sites by shared absences. For environmental data, a Euclidean distance would be more useful.

CODE
```
copepod.dis <- vegdist(TransCopepodData, method="bray")
```

Now let's look at the nMDS with labels for Community and take a note of the stress value (how well the ordination represents the data). A great stress value is <0.1, an OK stress value is between 0.1 and 0.2, and >0.2 is not so good. A stress greater than 0.3 is essentially arbitrary.

Next we plot the nMDS, labelling the sites with treatment.

CODE

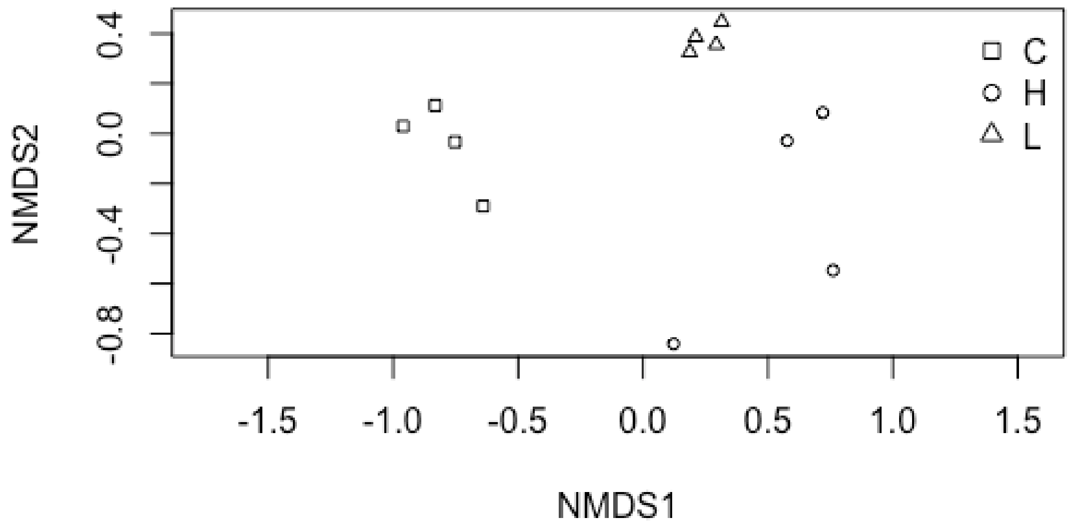
```
pchs<- c(0:2)
Copepod_Factor <- factor(CopepodData$Treatment)
nMDS_copepod <- metaMDS(TransCopepodData, distance="bray", k=2)
```

OUTPUT

```
Run 0 stress 0.1003895
Run 1 stress 0.1003895
... Procrustes: rmse 1.091751e-05  max resid 2.162003e-05
... Similar to previous best
Run 2 stress 0.1213244
Run 3 stress 0.121324
Run 4 stress 0.1102998
Run 5 stress 0.2209856
Run 6 stress 0.09313703
... New best solution
... Procrustes: rmse 0.06920781  max resid 0.1880591
Run 7 stress 0.1003895
Run 8 stress 0.1213242
Run 9 stress 0.1145475
Run 10 stress 0.1003895
Run 11 stress 0.1145492
Run 12 stress 0.1601729
Run 13 stress 0.09313703
... New best solution
... Procrustes: rmse 4.91324e-06  max resid 7.582151e-06
... Similar to previous best
Run 14 stress 0.09313703
... Procrustes: rmse 7.018053e-06  max resid 1.03343e-05
... Similar to previous best
Run 15 stress 0.110277
Run 16 stress 0.1213243
Run 17 stress 0.1213243
Run 18 stress 0.1003895
Run 19 stress 0.09313704
... Procrustes: rmse 3.431347e-05  max resid 5.199392e-05
... Similar to previous best
Run 20 stress 0.121324
*** Best solution repeated 3 times
```

CODE

```
plot_copepod <- ordiplot(nMDS_copepod, display = "sites", type="n")
points(nMDS_copepod, col="black", pch = pchs[Copepod_Factor])
legend("topright", bty = "n", legend = levels(Copepod_Factor), pch = pchs)
```



CODE

```
nMDS_copepod
```

OUTPUT

```

Call:
metaMDS(comm = TransCopepodData, distance = "bray", k = 2)

global Multidimensional Scaling using monoMDS

Data:     TransCopepodData
Distance: bray

Dimensions: 2
Stress:     0.09313703
Stress type 1, weak ties
Best solution was repeated 3 times in 20 tries
The best solution was from try 13 (random start)
Scaling: centring, PC rotation, halfchange scaling
Species: expanded scores based on 'TransCopepodData'
```

Exercise: Can you see a separation between treatments for copepod assemblages?

Exercise: What is the stress and is it good?

Let's test if the groups are significantly different using ANOSIM (Analysis of Similarities), a non-parametric permutation test. We will use 999 permutations to calculate the significance level.

So, let's test for Treatment

CODE

```r
copepod.anosim <- with(CopepodData, anosim(copepod.dis, CopepodData$Treatment))
summary(copepod.anosim)
```

OUTPUT

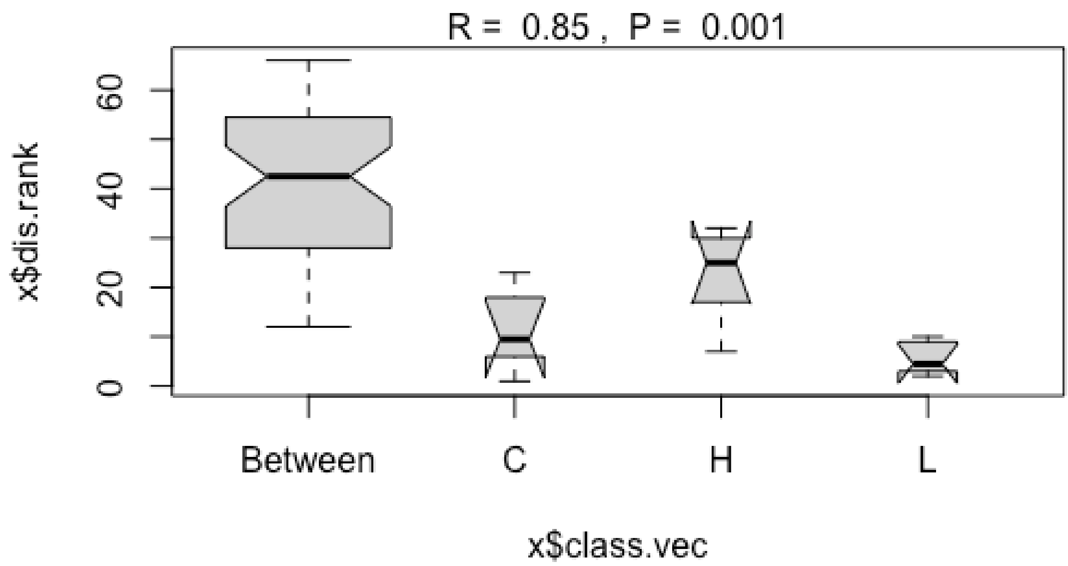
```

Call:
anosim(x = copepod.dis, grouping = CopepodData$Treatment)
Dissimilarity: bray

ANOSIM statistic R: 0.8495
      Significance: 0.001

Permutation: free
Number of permutations: 999

Upper quantiles of permutations (null model):
  90%   95% 97.5%   99%
0.213 0.273 0.332 0.463

Dissimilarity ranks between and within classes:
        0%   25%  50%   75% 100%  N
Between 12 28.50 42.5 54.25   66 48
C        1  6.50  9.5 16.25   23  6
H        7 18.25 25.0 29.50   32  6
L        2  3.25  4.5  8.00   10  6
```

CODE

```r
plot(copepod.anosim)
```

OUTPUT

```
Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, : some
notches went outside hinges ('box'): maybe set notch=FALSE
```



Is there a significant difference between the different treatments? We can see that using a brand-new ANOSIM code.

```
CODE
# Extract a grouping factor (e.g., Management type)
group <- CopepodData$Treatment

# Perform ANOSIM for all groups combined
anosim_all <- anosim(copepod.dis, group)
print(anosim_all)
```

```
OUTPUT

Call:
anosim(x = copepod.dis, grouping = group)
Dissimilarity: bray

ANOSIM statistic R: 0.8495
      Significance: 0.001

Permutation: free
Number of permutations: 999
```

```
CODE
pairwise.anosim <- function(copepod.dis, group, p.adjust.method = "holm", ...) {
  # Ensure grouping is a factor
  group <- factor(group)
  groups <- levels(group)

  # Generate all unique pairs of group levels
```

```r
  pairs <- t(combn(groups, 2))

  # Prepare storage for results
  results <- data.frame(
    group1  = character(),
    group2  = character(),
    R       = numeric(),
    p.value = numeric(),
    stringsAsFactors = FALSE
  )

  # Loop through each pair
  for (i in seq_len(nrow(pairs))) {
    g1 <- pairs[i, 1]
    g2 <- pairs[i, 2]

    # Subset the data to just these two groups
    keep <- group %in% c(g1, g2)
    dist_sub <- as.dist(as.matrix(copepod.dis)[keep, keep])
    group_sub <- droplevels(group[keep])

    # Run ANOSIM
    anosim_res <- anosim(dist_sub, group_sub, ...)

    # Store results
    results <- rbind(
      results,
      data.frame(
        group1  = g1,
        group2  = g2,
        R       = anosim_res$statistic,
        p.value = anosim_res$signif,
        stringsAsFactors = FALSE
      )
    )
  }

  # Adjust p-values for multiple comparisons
  results$p.adjusted <- p.adjust(results$p.value, method = p.adjust.method)

  return(results)
}

pairwise_results <- pairwise.anosim(copepod.dis, group, p.adjust.method = "holm")
pairwise_results
```

```
OUTPUT
  group1 group2       R p.value p.adjusted
1      C      H 0.96875   0.032      0.078
2      C      L 1.00000   0.029      0.078
3      H      L 0.59375   0.026      0.078
```

Exercise: Was there a significant difference between treatments for copepod assemblages? What was the Global R value? Were there significant differences between sites?

To see what species of copepods are contributing to the differences between treatments, we can use a SIMPER analysis (Similarity Percentages). The average is the amount that species contributes to the dissimilarity between the groups. The sd is the standard devation (i.e the variation) of the average dissimilarity contribution. The ratio is the average dissimilarity/sd.

The ava and avb are the average abundances of that species to the group. The cumsum is the ordered cumulative contribution to the dissimilarity between groups. Note we can ignore the cumulative 70%.

Note we use the ratio as the rank for the species that mostly separate the groups.

If R approaches 1 the dissimilarity between groups is greater than the dissimilarity within groups, if R is close to 0 the dissimilarity within groups is about the same as the dissimilarity between groups, and if R approaches −1, the variation within groups is greater than the dissimilarity between groups.

CODE

```
copepod.simper <- simper(TransCopepodData, CopepodData$Treatment)
summary(copepod.simper)
```

OUTPUT

```
Contrast: C_L

                average       sd    ratio      ava      avb cumsum     p
Tisbe.sp.4      0.12602  0.01742  7.23200  0.00000  3.64100  0.189 0.003 **
Tisbe.sp.2      0.08194  0.00696 11.77800  0.00000  2.37300  0.311 0.008 **
Tisbe.sp.3      0.07820  0.04708  1.66100  0.00000  2.28200  0.428 0.044 *
Tisbe.sp.5      0.05630  0.05839  0.96400  0.00000  1.66100  0.513 0.319
Halect.gothic   0.05107  0.02030  2.51600  0.50000  1.96800  0.589 0.052 .
Ameira.parvula  0.04409  0.00342 12.89100  0.00000  1.27700  0.655 0.017 *
Cyclopoida      0.02920  0.01798  1.62400  0.84460  0.00000  0.699 0.195
Copepodit.ind   0.02868  0.01653  1.73500  0.25000  1.07900  0.742 0.148
Stenhelia.refl  0.02527  0.01845  1.37000  0.82900  0.29700  0.780 0.419
Amphiascus.ten  0.02428  0.02342  1.03700  0.25000  0.68900  0.816 0.209
Proameira.simp  0.02367  0.02489  0.95100  0.00000  0.67100  0.852 0.009 **
Enhydros.long   0.02296  0.01856  1.23700  1.14190  0.57900  0.886 0.843
Ancorab.mirab   0.02015  0.02351  0.85700  1.03610  1.47000  0.916 0.876
Bulbamph.imus   0.01802  0.01877  0.96000  0.54730  0.00000  0.943 0.379
Daniel.fusifo   0.01324  0.00931  1.42200  1.00000  1.37300  0.963 0.917
Laophontidae    0.00891  0.01597  0.55800  0.00000  0.25000  0.976 0.015 *
Leptopsy.para   0.00873  0.01563  0.55800  0.25000  0.00000  0.990 0.669
Typhlam.typhl   0.00703  0.00550  1.27800  1.30170  1.47700  1.000 1.000
Tisbe.sp.1      0.00000  0.00000      NaN  0.00000  0.00000  1.000    NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: C_H

               average      sd   ratio     ava     avb cumsum     p
Tisbe.sp.1     0.12838 0.05504 2.33200 0.00000 2.28280  0.155 0.001 ***
Tisbe.sp.3     0.07987 0.05554 1.43800 0.00000 1.48270  0.251 0.064 .
Typhlam.typhl  0.07134 0.01048 6.80800 1.30170 0.00000  0.337 0.001 ***
Tisbe.sp.2     0.06684 0.04205 1.59000 0.00000 1.31490  0.418 0.064 .
Enhydros.long  0.06297 0.01168 5.39100 1.14190 0.00000  0.494 0.001 ***
Tisbe.sp.5     0.06165 0.04244 1.45300 0.00000 1.23230  0.568 0.168
Tisbe.sp.4     0.04869 0.05131 0.94900 0.00000 1.02410  0.627 0.991
Stenhelia.refl 0.04526 0.02867 1.57900 0.82900 0.00000  0.681 0.006 **
Ameira.parvula 0.04380 0.02667 1.64200 0.00000 0.84460  0.734 0.020 *
Ancorab.mirab  0.04305 0.03671 1.17300 1.03610 1.14890  0.786 0.088 .
Cyclopoida     0.03908 0.03005 1.30000 0.84460 0.25000  0.833 0.029 *
Daniel.fusifo  0.03472 0.02934 1.18300 1.00000 0.57900  0.875 0.023 *
Bulbamph.imus  0.02783 0.02938 0.94700 0.54730 0.00000  0.909 0.031 *
Halect.gothic  0.02726 0.02902 0.93900 0.50000 0.25000  0.942 0.998
Copepodit.ind  0.02182 0.02962 0.73700 0.25000 0.25000  0.968 0.661
```

```
Leptopsy.para  0.01395 0.02527 0.55200 0.25000 0.00000  0.985 0.111
Amphiascus.ten 0.01277 0.02308 0.55300 0.25000 0.00000  1.000 0.819
Proameira.simp 0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Laophontidae   0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: L_H

               average      sd   ratio     ava     avb cumsum     p
Tisbe.sp.4     0.08575 0.04324 1.98300 3.64100 1.02410  0.149 0.083 .
Tisbe.sp.1     0.07326 0.02796 2.62000 0.00000 2.28280  0.276 0.095 .
Halect.gothic  0.05378 0.01410 3.81500 1.96800 0.25000  0.370 0.022 *
Tisbe.sp.5     0.05238 0.03401 1.54000 1.66100 1.23230  0.460 0.493
Tisbe.sp.3     0.04921 0.03207 1.53500 2.28200 1.48270  0.546 0.900
Typhlam.typhl  0.04684 0.00580 8.07600 1.47700 0.00000  0.627 0.033 *
Tisbe.sp.2     0.03665 0.02956 1.24000 2.37300 1.31490  0.691 0.993
Daniel.fusifo  0.02801 0.02171 1.29000 1.37300 0.57900  0.740 0.196
Copepodit.ind  0.02569 0.01470 1.74700 1.07900 0.25000  0.784 0.275
Amphiascus.ten 0.02244 0.02375 0.94500 0.68900 0.00000  0.823 0.336
Ancorab.mirab  0.02243 0.01715 1.30700 1.47000 1.14890  0.862 0.837
Proameira.simp 0.02169 0.02288 0.94800 0.67100 0.00000  0.900 0.056 .
Enhydros.long  0.01780 0.01868 0.95300 0.57900 0.00000  0.930 0.989
Ameira.parvula 0.01457 0.01799 0.81000 1.27700 0.84460  0.956 0.999
Stenhelia.refl 0.00885 0.01589 0.55700 0.29700 0.00000  0.971 0.978
Cyclopoida     0.00843 0.01509 0.55800 0.00000 0.25000  0.986 0.995
Laophontidae   0.00816 0.01466 0.55700 0.25000 0.00000  1.000 0.170
Bulbamph.imus  0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Leptopsy.para  0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Permutation: free
Number of permutations: 999
```

Exercise: Which species most contribute to the separation between treatments?

Exercise 2: Analysis of ant assemblages from different Sydney Vegetation Communties

This is a study of the diversity of ants found in bushland remnants in the Sydney region. The ants were sampled with pitfall traps in the ground, and taken back to the lab to determine their species and relative abundance in each sample.

In this example we are looking at a collection of ants sampled in different vegetation communities ("Community") and whether they were sampled in the interior or on the edge of the vegetation community ("Sample"). We expected that the abundance and diversity of ants (the assemblage) would be different in different vegetation communities, and that the diversity of ants would be different in the interior to the edge of the vegetation community.

This leads to three multivariate null hypotheses.

H0: There is no difference in ant assemblage between the different vegetation communities.

H0: There is no difference in ant assemblage between the interior and the edge of the community.

H0: There is no interaction between the vegetation community and the location for the ant assemblages

We want to know if there is a pattern in species assemblages using nMDS, whether there is a statistically significant difference between communities and samples using PERMANOVA, and what species contribute to these differences via SIMPER.

First you need to load the packages vegan and MASS and file AntData.csv,

CODE

```
library(vegan)
library(MASS)

AntData <- read.csv("data/AntDataTotal.csv", header = TRUE)
```

Now the data is entered, let's apply a 4th root transformation, like in the example above,

CODE

```
TransAntData <- (AntData[,4:103])^(1/4)
```

Now let's generate a Bray-Curtis dissimilarity matrix on this transformed data.

CODE

```
ant.dis <- vegdist(TransAntData, method = "bray")
```

Now let's look at the nMDS with labels for Community and take a note of the stress value (how well the ordination represents the data). A great stress value is <0.1, an OK stress value is between 0.1 and 0.2, and >0.2 is not so good. A stress greater than 0.3 is essentially arbitrary.

CODE

```
pchs<- c(0:5)

Ant_Factor <- factor(AntData$Community)
nMDS_Ant <- metaMDS(TransAntData, distance="bray", k=2)
```

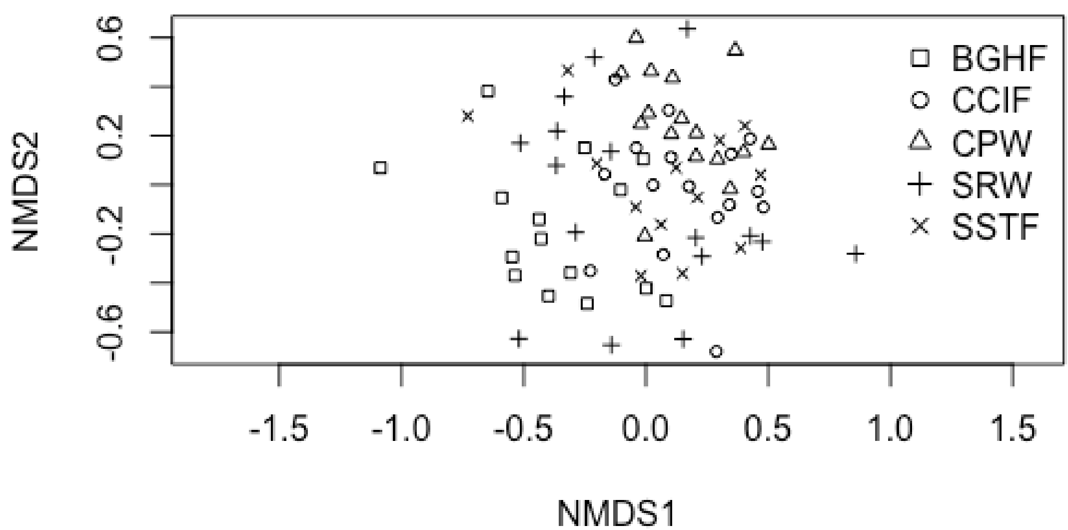
OUTPUT

```
Run 0 stress 0.2869097
Run 1 stress 0.2894127
Run 2 stress 0.28747
Run 3 stress 0.2900984
Run 4 stress 0.2885558
Run 5 stress 0.3023698
Run 6 stress 0.2887152
Run 7 stress 0.2915852
Run 8 stress 0.2952024
Run 9 stress 0.2995138
```

```
Run 10 stress 0.2880982
Run 11 stress 0.2879558
Run 12 stress 0.2949186
Run 13 stress 0.2987055
Run 14 stress 0.2985729
Run 15 stress 0.2874015
... Procrustes: rmse 0.05445269  max resid 0.2854394
Run 16 stress 0.2946961
Run 17 stress 0.2931475
Run 18 stress 0.2863242
... New best solution
... Procrustes: rmse 0.03371078  max resid 0.1651
Run 19 stress 0.2886178
Run 20 stress 0.2877279
*** Best solution was not repeated -- monoMDS stopping criteria:
     2: no. of iterations >= maxit
    18: stress ratio > sratmax
```

CODE

```
plot_Ant <- ordiplot(nMDS_Ant, display = "sites", type="n")
points(nMDS_Ant, col="black", pch = pchs[Ant_Factor])
legend("topright", bty = "n", legend = levels(Ant_Factor), pch = pchs)
```



CODE

```
nMDS_Ant
```

OUTPUT

```
Call:
metaMDS(comm = TransAntData, distance = "bray", k = 2)
```

```
global Multidimensional Scaling using monoMDS

Data:     TransAntData
Distance: bray

Dimensions: 2
Stress:     0.2863242
Stress type 1, weak ties
Best solution was not repeated after 20 tries
The best solution was from try 18 (random start)
Scaling: centring, PC rotation, halfchange scaling
Species: expanded scores based on 'TransAntData'
```

Exercise: Can you see a separation between communties?

Exercise: What is the stress and is it good?

We can test for differences between Communities, Samples and their Interactions using PERMANOVA (called Adonis in the package vegan). This is better for analyses with at least 2 factors than ANOSIM, as it can test the interactions. We use 999 permutations to calculate the significance (p) value. You interpret the PERMANOVA table like an ANOVA Table, except permutations are used to generate the p value,

CODE

```
# PERMANOVA with interaction using adonis2
adonis2(vegdist(TransAntData) ~ Community * Sample,
        data = AntData,
        permutations = 999,
        by = "terms")
```

OUTPUT

```
Permutation test for adonis under reduced model
Terms added sequentially (first to last)
Permutation: free
Number of permutations: 999

adonis2(formula = vegdist(TransAntData) ~ Community * Sample, data = AntData, permutations = 999, by =
"terms")
                 Df SumOfSqs      R2      F Pr(>F)
Community         4   2.8190 0.13512 2.6985  0.001 ***
Sample            1   0.1918 0.00919 0.7345  0.753
Community:Sample  4   0.6160 0.02953 0.5897  0.994
Residual         66  17.2367 0.82616
Total            75  20.8636 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Exercise: Is there a significant difference for the Community, Sample and/or their interaction?

If we find an overall significant p value, we need to see which level is significant from which other level. Here we can do pairwise tests (a bit like *post-hoc* tests in ANOVAs), with the p value adjusted with a Bonferroni correction. Here we need to create a function to do this, as it is not standard in the vegan package. So copy and paste this function.

To do the pairwise tests, there is a new R command on github and you can copy this text below to do it. This is for the Community factor in the ant data.

Note the first time you do this, you need to first install Rtools from here https://cran.r-project.org/bin/windows/Rtools/

CODE
```
library(devtools)
```

OUTPUT
```
Loading required package: usethis
```

OUTPUT
```

Attaching package: 'devtools'
```

OUTPUT
```
The following object is masked from 'package:permute':

    check
```

CODE
```
install_github("pmartinezarbizu/pairwiseAdonis/pairwiseAdonis")
```

OUTPUT
```
Warning: `install_github()` was deprecated in devtools 2.5.0.
ℹ Please use pak::pak("user/repo") instead.
```

OUTPUT
```
Using GitHub PAT from the git credential store.
```

OUTPUT
```
Skipping install of 'pairwiseAdonis' from a github remote, the SHA1 (cb190f76) has not changed since
last install.
  Use `force = TRUE` to force installation
```

CODE
```
library(pairwiseAdonis)
```

OUTPUT
```
Loading required package: cluster
```

CODE
```
pairwise.adonis(TransAntData, AntData$Community)
```

OUTPUT
```
          pairs Df SumsOfSqs  F.Model         R2 p.value p.adjusted sig
1  BGHF vs CCIF  1 1.1326581 4.532465 0.13516648   0.001       0.01   *
```

```
2   BGHF vs CPW  1 1.3163257 5.351814 0.15579421   0.001       0.01   *
3   BGHF vs SRW  1 0.6422157 2.204225 0.07063867   0.007       0.07
4  BGHF vs SSTF  1 0.7759586 2.965274 0.10237341   0.001       0.01   *
5   CCIF vs CPW  1 0.4603595 2.075474 0.06470596   0.018       0.18
6   CCIF vs SRW  1 0.5712089 2.149875 0.06687041   0.005       0.05   .
7  CCIF vs SSTF  1 0.3236528 1.381556 0.04867796   0.154       1.00
8    CPW vs SRW  1 0.8702832 3.323148 0.09972490   0.001       0.01   *
9   CPW vs SSTF  1 0.4584616 1.993009 0.06874103   0.017       0.17
10  SRW vs SSTF  1 0.4534517 1.626466 0.05681688   0.087       0.87
```

Exercise: Which communities are significantly different from each other?

To see what species are contributing to the differences between the vegetation communities, we can again use a SIMPER analysis (Similarity Percentages)

CODE

```
antsim <- simper(TransAntData, AntData$Community)
summary(antsim)
```

OUTPUT

```
Contrast: BGHF_CCIF

                          average      sd   ratio     ava     avb cumsum     p
Pheidole.5                0.05345 0.04054 1.31860 0.44630 1.22750  0.069 0.001
Tetramorium.3             0.04337 0.03735 1.16130 0.71720 0.00000  0.125 0.001
Anonychromyrma.1          0.03847 0.03475 1.10700 0.69070 0.25380  0.174 0.019
Pheidole.7                0.03459 0.03500 0.98840 0.50690 0.38970  0.219 0.032
Rhytidiponera..metallica. 0.03448 0.03747 0.92030 0.98380 1.30250  0.263 0.308
Tapinoma.1                0.03402 0.03186 1.06780 0.42520 0.65120  0.307 0.125
Monomorium.1              0.03233 0.04049 0.79850 0.45680 0.29450  0.349 0.057
Meranoplus.1              0.02743 0.03296 0.83210 0.34180 0.33620  0.384 0.466
Notoncus.1                0.02495 0.03250 0.76780 0.20000 0.37000  0.416 0.702
Paratrechina.1            0.02038 0.02882 0.70720 0.21260 0.25000  0.443 0.846
Machomyrma.1              0.01886 0.03016 0.62530 0.00000 0.33620  0.467 0.605
Paratrechina.4            0.01802 0.02650 0.68000 0.26670 0.14480  0.490 0.114
Pheidole.7.1              0.01604 0.02829 0.56710 0.30690 0.00000  0.511 0.110
Heteroponera.1            0.01543 0.02693 0.57290 0.13330 0.18750  0.531 0.089
Tetramorium.4             0.01536 0.02604 0.58980 0.30040 0.00000  0.550 0.002
Iridomyrmex.7             0.01409 0.02534 0.55610 0.00000 0.26180  0.569 0.002
Iridomyrmex.purpureus     0.01395 0.02541 0.54900 0.00000 0.26180  0.587 0.847
Paratrechina.2            0.01262 0.02405 0.52460 0.15440 0.12500  0.603 0.928
Solenopsis.1              0.01200 0.02462 0.48760 0.20000 0.00000  0.618 0.033
Polyrachis.5              0.01151 0.02474 0.46500 0.00000 0.18750  0.633 0.007
Machomyrma.3              0.01131 0.02500 0.45240 0.00000 0.18750  0.648 0.120
Doleromyrma.1             0.01120 0.02414 0.46390 0.00000 0.18750  0.662 0.653
Pristomyrmex.1            0.01119 0.02335 0.47930 0.21260 0.00000  0.676 0.048
Crematogaster.1           0.01117 0.02270 0.49230 0.20000 0.00000  0.691 0.922
Pristomyrmex.2            0.01080 0.02361 0.45740 0.14590 0.07430  0.705 0.786
Colobostruma.1            0.01019 0.02101 0.48500 0.20000 0.00000  0.718 0.005
Crematogaster.3           0.00987 0.02758 0.35770 0.00000 0.16760  0.731 0.054
Rhopalomastix.2           0.00920 0.02555 0.36020 0.09430 0.06250  0.743 0.081
Camponotus.12             0.00920 0.01972 0.46660 0.00000 0.19930  0.754 0.596
Pheidole.6                0.00919 0.01970 0.46670 0.00000 0.19930  0.766 0.784
Rhopalomastix.1           0.00909 0.02516 0.36140 0.00000 0.14480  0.778 0.204
Rhopalomastix.3           0.00792 0.02083 0.38030 0.15860 0.00000  0.788 0.017
Notoncus.4                0.00767 0.02079 0.36910 0.00000 0.13680  0.798 0.065
Myrmecia.1                0.00689 0.01809 0.38080 0.13330 0.00000  0.807 0.100
Camponotus.consobrinus    0.00682 0.01929 0.35360 0.00000 0.12500  0.816 0.896
Iridomyrmex.2             0.00680 0.01856 0.36640 0.00000 0.12500  0.825 0.822
Myrmecia.3                0.00677 0.01784 0.37920 0.13330 0.00000  0.833 0.518
```

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Mayriella.2              0.00669 0.01851 0.36130 0.06670 0.06250  0.842 0.773
Prolasius.1              0.00666 0.01822 0.36530 0.07930 0.06250  0.850 0.280
Polyrachis.2             0.00624 0.01709 0.36490 0.06670 0.06250  0.859 0.226
Ochetellus.1             0.00618 0.01701 0.36340 0.00000 0.12500  0.866 0.934
Doleromyrma.3            0.00608 0.01575 0.38600 0.13330 0.00000  0.874 0.889
Strumigenys.1            0.00580 0.01599 0.36290 0.06670 0.06250  0.882 0.294
Papyrius.1               0.00559 0.01535 0.36420 0.06670 0.06250  0.889 0.217
Paratrechina.6           0.00434 0.01644 0.26400 0.07930 0.00000  0.894 0.099
Prolasius.2              0.00434 0.01644 0.26400 0.07930 0.00000  0.900 0.203
Crematogaster.2          0.00421 0.01598 0.26320 0.06670 0.00000  0.905 0.967
Doleromyrma.2            0.00401 0.01600 0.25070 0.00000 0.06250  0.911 0.227
Strumigenys.2            0.00400 0.01517 0.26350 0.06670 0.00000  0.916 0.108
Iridomyrmex.4            0.00388 0.01546 0.25120 0.00000 0.06250  0.921 0.667
Polyrachis.1             0.00388 0.01546 0.25120 0.00000 0.06250  0.926 0.206
Plagiolepis.1            0.00365 0.01382 0.26400 0.06670 0.00000  0.931 0.099
Melophorus.1             0.00365 0.01446 0.25210 0.00000 0.06250  0.935 0.227
Notoncus.3               0.00354 0.01340 0.26420 0.06670 0.00000  0.940 0.125
Prolasius.6              0.00354 0.01340 0.26420 0.06670 0.00000  0.944 0.938
Probolomyrmex.1          0.00339 0.01283 0.26440 0.06670 0.00000  0.949 0.101
Tetramorium.5            0.00339 0.01283 0.26440 0.06670 0.00000  0.953 0.101
Pheidole.2               0.00337 0.01274 0.26440 0.06670 0.00000  0.958 0.109
Prolasius.3              0.00337 0.01274 0.26440 0.06670 0.00000  0.962 0.661
Leptomyrmex.1            0.00335 0.01265 0.26440 0.06670 0.00000  0.966 0.109
Froggattella.1           0.00328 0.01296 0.25330 0.00000 0.06250  0.970 0.215
Colobostruma.2           0.00301 0.01186 0.25410 0.00000 0.06250  0.974 0.250
Machomyrma.6             0.00301 0.01183 0.25410 0.00000 0.06250  0.978 0.258
Melophorus.2             0.00301 0.01183 0.25410 0.00000 0.06250  0.982 0.340
Polyrachis.3             0.00301 0.01183 0.25410 0.00000 0.06250  0.986 0.703
Mayriella.1              0.00286 0.01078 0.26500 0.06670 0.00000  0.990 0.485
Papyrius.2               0.00286 0.01078 0.26500 0.06670 0.00000  0.993 0.963
Mesostruma.1             0.00273 0.01031 0.26520 0.06670 0.00000  0.997 0.973
Stigmacros.3             0.00253 0.00993 0.25520 0.00000 0.06250  1.000 0.853
Anonychromyrma.2         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Anonychromyrma.3         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Camponotis.26            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Camponotus.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Disturbed.lost           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Dolichonderus.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Epopostruma.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.5            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.6            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.8            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Machomyrma.4             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Meranoplus.2             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.2               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.4               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecorhrynchus.1       0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Oligomyrmex.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Oligomyrmex.2            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pachychondyla.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pachycondyla.2           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Paratrechina.5           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.1               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.6.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.7             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.8             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.4              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.7              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.5          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Stigmacros.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Stigmacros.2             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Stigmacros.4             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Technomyrmex.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA

Pheidole.5               ***
```

```
Tetramorium.3             ***
Anonychromyrma.1          *
Pheidole.7                *
Rhytidiponera..metallica.
Tapinoma.1
Monomorium.1              .
Meranoplus.1
Notoncus.1
Paratrechina.1
Machomyrma.1
Paratrechina.4
Pheidole.7.1
Heteroponera.1            .
Tetramorium.4             **
Iridomyrmex.7             **
Iridomyrmex.purpureus
Paratrechina.2
Solenopsis.1              *
Polyrachis.5              **
Machomyrma.3
Doleromyrma.1
Pristomyrmex.1            *
Crematogaster.1
Pristomyrmex.2
Colobostruma.1            **
Crematogaster.3           .
Rhopalomastix.2           .
Camponotus.12
Pheidole.6
Rhopalomastix.1
Rhopalomastix.3           *
Notoncus.4                .
Myrmecia.1                .
Camponotus.consobrinus
Iridomyrmex.2
Myrmecia.3
Mayriella.2
Prolasius.1
Polyrachis.2
Ochetellus.1
Doleromyrma.3
Strumigenys.1
Papyrius.1
Paratrechina.6            .
Prolasius.2
Crematogaster.2
Doleromyrma.2
Strumigenys.2
Iridomyrmex.4
Polyrachis.1
Plagiolepis.1             .
Melophorus.1
Notoncus.3
Prolasius.6
Probolomyrmex.1
Tetramorium.5
Pheidole.2
Prolasius.3
Leptomyrmex.1
Froggattella.1
Colobostruma.2
Machomyrma.6
Melophorus.2
Polyrachis.3
Mayriella.1
```

```
Papyrius.2
Mesostruma.1
Stigmacros.3
Anonychromyrma.2
Anonychromyrma.3
Camponotis.26
Camponotus.1
Disturbed.lost
Dolichonderus.1
Epopostruma.1
Iridomyrmex.5
Iridomyrmex.6
Iridomyrmex.8
Machomyrma.4
Meranoplus.2
Myrmecia.2
Myrmecia.4
Myrmecorhrynchus.1
Oligomyrmex.1
Oligomyrmex.2
Pachychondyla.1
Pachycondyla.2
Paratrechina.5
Pheidole.1
Pheidole.6.1
Polyrachis.7
Polyrachis.8
Prolasius.4
Prolasius.7
Rhopalomastix.5
Stigmacros.1
Stigmacros.2
Stigmacros.4
Technomyrmex.1
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: BGHF_CPW

                          average      sd   ratio     ava     avb cumsum     p
Tetramorium.3             0.03724 0.03269 1.13910 0.71720 0.00000  0.047 0.004
Pheidole.5                0.03715 0.03141 1.18270 0.44630 0.92190  0.094 0.595
Meranoplus.1              0.03196 0.02890 1.10590 0.34180 0.65660  0.135 0.089
Anonychromyrma.1          0.03196 0.02791 1.14510 0.69070 0.08230  0.176 0.310
Tapinoma.1                0.03127 0.02931 1.06680 0.42520 0.73330  0.215 0.446
Rhytidiponera..metallica. 0.03040 0.03400 0.89420 0.98380 1.45860  0.254 0.586
Pheidole.7                0.02792 0.03003 0.92990 0.50690 0.34410  0.290 0.472
Paratrechina.1            0.02761 0.02785 0.99110 0.21260 0.57810  0.325 0.148
Iridomyrmex.purpureus     0.02585 0.03202 0.80740 0.00000 0.54280  0.357 0.059
Monomorium.1              0.02474 0.03321 0.74510 0.45680 0.16760  0.389 0.432
Notoncus.1                0.02287 0.02804 0.81560 0.20000 0.42060  0.418 0.854
Machomyrma.1              0.02194 0.02674 0.82050 0.00000 0.43750  0.446 0.328
Pheidole.7.1              0.01935 0.02739 0.70640 0.30690 0.19930  0.470 0.015
Mesostruma.1              0.01824 0.02621 0.69570 0.06670 0.39440  0.494 0.005
Crematogaster.1           0.01707 0.02408 0.70900 0.20000 0.25000  0.515 0.550
Iridomyrmex.5             0.01685 0.02711 0.62160 0.00000 0.34800  0.537 0.001
Paratrechina.2            0.01512 0.02293 0.65940 0.15440 0.25000  0.556 0.838
Pristomyrmex.2            0.01511 0.02610 0.57890 0.14590 0.22520  0.575 0.479
Prolasius.6               0.01482 0.02421 0.61210 0.06670 0.31540  0.594 0.050
Papyrius.2                0.01478 0.02913 0.50740 0.06670 0.21120  0.613 0.144
Tetramorium.4             0.01448 0.02315 0.62550 0.30040 0.06250  0.631 0.009
Ochetellus.1              0.01402 0.02158 0.64950 0.00000 0.32430  0.649 0.272
Camponotus.consobrinus    0.01372 0.02459 0.55770 0.00000 0.25000  0.666 0.373
Mayriella.2               0.01347 0.02244 0.60030 0.06670 0.25000  0.683 0.141
Iridomyrmex.2             0.01314 0.02377 0.55280 0.00000 0.27370  0.700 0.304
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Paratrechina.4          0.01304 0.02090 0.62380 0.26670 0.06250  0.717 0.501
Camponotus.12           0.01088 0.01960 0.55510 0.00000 0.25000  0.730 0.414
Solenopsis.1            0.01034 0.02144 0.48210 0.20000 0.00000  0.744 0.116
Pristomyrmex.1          0.00978 0.02056 0.47570 0.21260 0.00000  0.756 0.131
Doleromyrma.3           0.00941 0.01784 0.52750 0.13330 0.12500  0.768 0.730
Stigmacros.3            0.00913 0.01969 0.46340 0.00000 0.21910  0.779 0.111
Colobostruma.1          0.00893 0.01855 0.48140 0.20000 0.00000  0.791 0.043
Doleromyrma.1           0.00841 0.01800 0.46710 0.00000 0.19930  0.802 0.850
Pheidole.6              0.00817 0.01769 0.46170 0.00000 0.18750  0.812 0.864
Myrmecia.3              0.00803 0.01780 0.45140 0.13330 0.06250  0.822 0.381
Heteroponera.1          0.00802 0.01758 0.45610 0.13330 0.07430  0.832 0.723
Myrmecia.1              0.00757 0.01689 0.44780 0.13330 0.06250  0.842 0.059
Polyrachis.2            0.00715 0.01591 0.44920 0.06670 0.12500  0.851 0.166
Rhopalomastix.3         0.00695 0.01840 0.37790 0.15860 0.00000  0.860 0.144
Polyrachis.8            0.00636 0.01715 0.37080 0.00000 0.12500  0.868 0.054
Prolasius.1             0.00549 0.01529 0.35900 0.07930 0.06250  0.875 0.475
Pachycondyla.2          0.00543 0.01485 0.36570 0.00000 0.12500  0.882 0.500
Mayriella.1             0.00541 0.01505 0.35970 0.06670 0.06250  0.889 0.161
Papyrius.1              0.00540 0.01484 0.36400 0.06670 0.06250  0.895 0.262
Polyrachis.3            0.00531 0.01445 0.36740 0.00000 0.12500  0.902 0.430
Rhopalomastix.2         0.00510 0.01960 0.26010 0.09430 0.00000  0.909 0.404
Iridomyrmex.4           0.00449 0.01767 0.25410 0.00000 0.09350  0.914 0.602
Paratrechina.6          0.00378 0.01446 0.26150 0.07930 0.00000  0.919 0.420
Prolasius.2             0.00378 0.01446 0.26150 0.07930 0.00000  0.924 0.423
Crematogaster.2         0.00360 0.01386 0.26010 0.06670 0.00000  0.929 0.980
Strumigenys.2           0.00345 0.01323 0.26060 0.06670 0.00000  0.933 0.368
Myrmecia.2              0.00344 0.01360 0.25280 0.00000 0.06250  0.937 0.242
Oligomyrmex.1           0.00319 0.01258 0.25360 0.00000 0.06250  0.942 0.226
Plagiolepis.1           0.00318 0.01216 0.26150 0.06670 0.00000  0.946 0.420
Notoncus.3              0.00309 0.01182 0.26170 0.06670 0.00000  0.949 0.407
Stigmacros.4            0.00300 0.01182 0.25410 0.00000 0.06250  0.953 0.570
Probolomyrmex.1         0.00298 0.01137 0.26200 0.06670 0.00000  0.957 0.406
Tetramorium.5           0.00298 0.01137 0.26200 0.06670 0.00000  0.961 0.406
Iridomyrmex.6           0.00297 0.01167 0.25420 0.00000 0.06250  0.965 0.228
Pheidole.2              0.00296 0.01130 0.26210 0.06670 0.00000  0.968 0.370
Prolasius.3             0.00296 0.01130 0.26210 0.06670 0.00000  0.972 0.821
Leptomyrmex.1           0.00294 0.01123 0.26210 0.06670 0.00000  0.976 0.383
Meranoplus.2            0.00278 0.01092 0.25470 0.00000 0.06250  0.979 0.282
Rhopalomastix.1         0.00267 0.01044 0.25580 0.00000 0.07430  0.983 0.859
Strumigenys.1           0.00255 0.00970 0.26320 0.06670 0.00000  0.986 0.801
Melophorus.2            0.00244 0.00954 0.25610 0.00000 0.07430  0.989 0.576
Pheidole.6.1            0.00241 0.00942 0.25550 0.00000 0.06250  0.992 0.285
Iridomyrmex.8           0.00205 0.00802 0.25610 0.00000 0.06250  0.995 0.277
Myrmecia.4              0.00205 0.00802 0.25610 0.00000 0.06250  0.997 0.277
Rhopalomastix.5         0.00205 0.00802 0.25610 0.00000 0.06250  1.000 0.277
Anonychromyrma.2        0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Anonychromyrma.3        0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Camponotis.26           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Camponotus.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Colobostruma.2          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Crematogaster.3         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Disturbed.lost          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Doleromyrma.2           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Dolichonderus.1         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Epopostruma.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Froggattella.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.7           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Machomyrma.3            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Machomyrma.4            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Machomyrma.6            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Melophorus.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecorhrynchus.1      0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Notoncus.4              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Oligomyrmex.2           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pachychondyla.1         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
```

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Paratrechina.5            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.1                0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.1              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.5              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.7              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.4               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.7               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Stigmacros.1              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Stigmacros.2              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Technomyrmex.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA

Tetramorium.3             **
Pheidole.5
Meranoplus.1              .
Anonychromyrma.1
Tapinoma.1
Rhytidiponera..metallica.
Pheidole.7
Paratrechina.1
Iridomyrmex.purpureus     .
Monomorium.1
Notoncus.1
Machomyrma.1
Pheidole.7.1              *
Mesostruma.1              **
Crematogaster.1
Iridomyrmex.5             ***
Paratrechina.2
Pristomyrmex.2
Prolasius.6               *
Papyrius.2
Tetramorium.4             **
Ochetellus.1
Camponotus.consobrinus
Mayriella.2
Iridomyrmex.2
Paratrechina.4
Camponotus.12
Solenopsis.1
Pristomyrmex.1
Doleromyrma.3
Stigmacros.3
Colobostruma.1            *
Doleromyrma.1
Pheidole.6
Myrmecia.3
Heteroponera.1
Myrmecia.1                .
Polyrachis.2
Rhopalomastix.3
Polyrachis.8              .
Prolasius.1
Pachycondyla.2
Mayriella.1
Papyrius.1
Polyrachis.3
Rhopalomastix.2
Iridomyrmex.4
Paratrechina.6
Prolasius.2
Crematogaster.2
Strumigenys.2
Myrmecia.2
Oligomyrmex.1
Plagiolepis.1
```

```
Notoncus.3
Stigmacros.4
Probolomyrmex.1
Tetramorium.5
Iridomyrmex.6
Pheidole.2
Prolasius.3
Leptomyrmex.1
Meranoplus.2
Rhopalomastix.1
Strumigenys.1
Melophorus.2
Pheidole.6.1
Iridomyrmex.8
Myrmecia.4
Rhopalomastix.5
Anonychromyrma.2
Anonychromyrma.3
Camponotis.26
Camponotus.1
Colobostruma.2
Crematogaster.3
Disturbed.lost
Doleromyrma.2
Dolichonderus.1
Epopostruma.1
Froggattella.1
Iridomyrmex.7
Machomyrma.3
Machomyrma.4
Machomyrma.6
Melophorus.1
Myrmecorhrynchus.1
Notoncus.4
Oligomyrmex.2
Pachychondyla.1
Paratrechina.5
Pheidole.1
Polyrachis.1
Polyrachis.5
Polyrachis.7
Prolasius.4
Prolasius.7
Stigmacros.1
Stigmacros.2
Technomyrmex.1
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: BGHF_SRW

                          average      sd   ratio     ava     avb cumsum     p
Tetramorium.3             0.04571 0.04213 1.08490 0.71720 0.29710  0.058 0.001
Pheidole.5                0.04554 0.04637 0.98200 0.44630 0.75730  0.117 0.015
Rhytidiponera..metallica. 0.04178 0.03930 1.06300 0.98380 0.79590  0.170 0.022
Anonychromyrma.1          0.03930 0.03593 1.09370 0.69070 0.35530  0.220 0.011
Pheidole.7                0.03562 0.04061 0.87700 0.50690 0.26180  0.266 0.013
Monomorium.1              0.03444 0.04336 0.79440 0.45680 0.27240  0.310 0.020
Notoncus.1                0.02975 0.03307 0.89970 0.20000 0.48480  0.348 0.234
Crematogaster.2           0.02843 0.03727 0.76300 0.06670 0.43640  0.384 0.002
Tapinoma.1                0.02781 0.03265 0.85190 0.42520 0.18750  0.420 0.894
Doleromyrma.3             0.02663 0.04353 0.61190 0.13330 0.33670  0.454 0.003
Meranoplus.1              0.02641 0.03671 0.71920 0.34180 0.24430  0.487 0.579
Paratrechina.1            0.02429 0.03114 0.78010 0.21260 0.34410  0.518 0.507
Paratrechina.2            0.02210 0.02986 0.74040 0.15440 0.33230  0.547 0.241
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Paratrechina.4           0.01956 0.02986 0.65500 0.26670 0.12500  0.572 0.047
Pheidole.7.1             0.01918 0.03155 0.60790 0.30690 0.06250  0.596 0.021
Pristomyrmex.2           0.01825 0.03129 0.58320 0.14590 0.21340  0.619 0.211
Tetramorium.4            0.01672 0.02853 0.58580 0.30040 0.00000  0.641 0.001
Solenopsis.1             0.01533 0.02859 0.53620 0.20000 0.06250  0.660 0.001
Crematogaster.1          0.01470 0.02711 0.54230 0.20000 0.06250  0.679 0.739
Myrmecia.3               0.01374 0.02702 0.50830 0.13330 0.12500  0.697 0.019
Heteroponera.1           0.01289 0.02495 0.51650 0.13330 0.12500  0.713 0.281
Pristomyrmex.1           0.01222 0.02573 0.47480 0.21260 0.00000  0.729 0.023
Colobostruma.1           0.01110 0.02310 0.48050 0.20000 0.00000  0.743 0.001
Prolasius.3              0.01032 0.02298 0.44900 0.06670 0.13680  0.756 0.045
Iridomyrmex.2            0.00923 0.02626 0.35130 0.00000 0.12500  0.768 0.672
Rhopalomastix.3          0.00862 0.02285 0.37710 0.15860 0.00000  0.779 0.002
Technomyrmex.1           0.00828 0.02270 0.36470 0.00000 0.12500  0.789 0.041
Pachycondyla.2           0.00810 0.02202 0.36800 0.00000 0.13680  0.800 0.207
Prolasius.1              0.00767 0.02104 0.36470 0.07930 0.06250  0.810 0.189
Mayriella.2              0.00759 0.02110 0.35990 0.06670 0.06250  0.819 0.692
Prolasius.2              0.00758 0.02102 0.36080 0.07930 0.06250  0.829 0.028
Myrmecia.1               0.00751 0.01991 0.37740 0.13330 0.00000  0.839 0.064
Dolichonderus.1          0.00722 0.01961 0.36820 0.00000 0.12500  0.848 0.037
Machomyrma.3             0.00707 0.01917 0.36910 0.00000 0.12500  0.857 0.494
Stigmacros.1             0.00692 0.01872 0.36950 0.00000 0.12500  0.866 0.186
Rhopalomastix.2          0.00661 0.02531 0.26120 0.09430 0.00000  0.874 0.147
Papyrius.2               0.00623 0.01722 0.36180 0.06670 0.06250  0.882 0.827
Strumigenys.1            0.00623 0.01722 0.36180 0.06670 0.06250  0.890 0.197
Paratrechina.6           0.00475 0.01806 0.26270 0.07930 0.00000  0.896 0.003
Strumigenys.2            0.00442 0.01687 0.26180 0.06670 0.00000  0.902 0.003
Iridomyrmex.purpureus    0.00434 0.01743 0.24920 0.00000 0.06250  0.907 0.996
Plagiolepis.1            0.00399 0.01519 0.26270 0.06670 0.00000  0.912 0.003
Notoncus.3               0.00386 0.01467 0.26300 0.06670 0.00000  0.917 0.001
Polyrachis.2             0.00386 0.01467 0.26300 0.06670 0.00000  0.922 0.545
Prolasius.6              0.00386 0.01467 0.26300 0.06670 0.00000  0.927 0.879
Probolomyrmex.1          0.00368 0.01398 0.26330 0.06670 0.00000  0.932 0.001
Tetramorium.5            0.00368 0.01398 0.26330 0.06670 0.00000  0.937 0.001
Papyrius.1               0.00365 0.01387 0.26340 0.06670 0.00000  0.941 0.367
Pheidole.2               0.00365 0.01387 0.26340 0.06670 0.00000  0.946 0.002
Leptomyrmex.1            0.00363 0.01377 0.26340 0.06670 0.00000  0.951 0.003
Machomyrma.1             0.00357 0.01415 0.25240 0.00000 0.06250  0.955 1.000
Camponotus.consobrinus   0.00353 0.01397 0.25250 0.00000 0.06250  0.960 0.985
Doleromyrma.1            0.00353 0.01397 0.25250 0.00000 0.06250  0.964 0.974
Iridomyrmex.4            0.00350 0.01387 0.25260 0.00000 0.06250  0.969 0.745
Pachychondyla.1          0.00345 0.01365 0.25280 0.00000 0.06250  0.973 0.228
Rhopalomastix.1          0.00341 0.01350 0.25290 0.00000 0.06250  0.977 0.754
Stigmacros.4             0.00325 0.01284 0.25340 0.00000 0.06250  0.982 0.325
Mayriella.1              0.00306 0.01156 0.26440 0.06670 0.00000  0.986 0.359
Mesostruma.1             0.00292 0.01102 0.26460 0.06670 0.00000  0.989 0.949
Ochetellus.1             0.00282 0.01107 0.25460 0.00000 0.06250  0.993 0.986
Oligomyrmex.2            0.00282 0.01107 0.25460 0.00000 0.06250  0.996 0.503
Pheidole.6               0.00282 0.01107 0.25460 0.00000 0.06250  1.000 0.993
Anonychromyrma.2         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Anonychromyrma.3         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Camponotis.26            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Camponotus.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Camponotus.12            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Colobostruma.2           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Crematogaster.3          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Disturbed.lost           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Doleromyrma.2            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Epopostruma.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Froggattella.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.5            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.6            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.7            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.8            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Machomyrma.4             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
```

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Machomyrma.6              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Melophorus.1              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Melophorus.2              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Meranoplus.2              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.2                0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.4                0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecorhrynchus.1        0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Notoncus.4                0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Oligomyrmex.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Paratrechina.5            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.1                0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.6.1              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.1              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.3              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.5              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.7              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.8              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.4               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.7               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.5           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Stigmacros.2              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Stigmacros.3              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA

Tetramorium.3             ***
Pheidole.5                *
Rhytidiponera..metallica. *
Anonychromyrma.1          *
Pheidole.7                *
Monomorium.1              *
Notoncus.1
Crematogaster.2           **
Tapinoma.1
Doleromyrma.3             **
Meranoplus.1
Paratrechina.1
Paratrechina.2
Paratrechina.4            *
Pheidole.7.1              *
Pristomyrmex.2
Tetramorium.4             ***
Solenopsis.1              ***
Crematogaster.1
Myrmecia.3                *
Heteroponera.1
Pristomyrmex.1            *
Colobostruma.1            ***
Prolasius.3               *
Iridomyrmex.2
Rhopalomastix.3           **
Technomyrmex.1            *
Pachycondyla.2
Prolasius.1
Mayriella.2
Prolasius.2               *
Myrmecia.1                .
Dolichonderus.1           *
Machomyrma.3
Stigmacros.1
Rhopalomastix.2
Papyrius.2
Strumigenys.1
Paratrechina.6            **
Strumigenys.2             **
Iridomyrmex.purpureus
Plagiolepis.1             **
```

```
Notoncus.3                ***
Polyrachis.2
Prolasius.6
Probolomyrmex.1           ***
Tetramorium.5             ***
Papyrius.1
Pheidole.2                **
Leptomyrmex.1             **
Machomyrma.1
Camponotus.consobrinus
Doleromyrma.1
Iridomyrmex.4
Pachychondyla.1
Rhopalomastix.1
Stigmacros.4
Mayriella.1
Mesostruma.1
Ochetellus.1
Oligomyrmex.2
Pheidole.6
Anonychromyrma.2
Anonychromyrma.3
Camponotis.26
Camponotus.1
Camponotus.12
Colobostruma.2
Crematogaster.3
Disturbed.lost
Doleromyrma.2
Epopostruma.1
Froggattella.1
Iridomyrmex.5
Iridomyrmex.6
Iridomyrmex.7
Iridomyrmex.8
Machomyrma.4
Machomyrma.6
Melophorus.1
Melophorus.2
Meranoplus.2
Myrmecia.2
Myrmecia.4
Myrmecorhrynchus.1
Notoncus.4
Oligomyrmex.1
Paratrechina.5
Pheidole.1
Pheidole.6.1
Polyrachis.1
Polyrachis.3
Polyrachis.5
Polyrachis.7
Polyrachis.8
Prolasius.4
Prolasius.7
Rhopalomastix.5
Stigmacros.2
Stigmacros.3
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: BGHF_SSTF

                          average      sd   ratio     ava     avb cumsum     p
Pheidole.5                0.04728 0.03956 1.19510 0.44630 1.08140  0.062 0.012
```

```
Tetramorium.3             0.04012 0.03587 1.11840 0.71720 0.25510  0.114 0.001
Anonychromyrma.1          0.03661 0.03367 1.08740 0.69070 0.49420  0.162 0.057
Rhytidiponera..metallica. 0.03363 0.03803 0.88450 0.98380 1.22640  0.207 0.374
Pheidole.7                0.03164 0.03434 0.92150 0.50690 0.24530  0.248 0.159
Crematogaster.1           0.03090 0.03234 0.95540 0.20000 0.56960  0.289 0.002
Monomorium.1              0.02876 0.03777 0.76140 0.45680 0.20030  0.326 0.191
Tapinoma.1                0.02771 0.02968 0.93340 0.42520 0.38460  0.363 0.877
Meranoplus.1              0.02179 0.03107 0.70120 0.34180 0.15380  0.391 0.854
Paratrechina.2            0.02106 0.02933 0.71790 0.15440 0.32220  0.419 0.371
Machomyrma.1              0.01990 0.02632 0.75610 0.00000 0.41370  0.445 0.547
Notoncus.1                0.01961 0.02981 0.65790 0.20000 0.24530  0.470 0.951
Pheidole.6                0.01656 0.02603 0.63620 0.00000 0.32220  0.492 0.153
Paratrechina.4            0.01614 0.02494 0.64730 0.26670 0.09150  0.513 0.243
Paratrechina.1            0.01610 0.02596 0.62000 0.21260 0.15380  0.534 0.963
Pheidole.7.1              0.01597 0.02834 0.56370 0.30690 0.00000  0.555 0.129
Tetramorium.4             0.01529 0.02607 0.58650 0.30040 0.00000  0.575 0.010
Pristomyrmex.2            0.01409 0.02577 0.54690 0.14590 0.15380  0.594 0.535
Pristomyrmex.1            0.01377 0.02511 0.54850 0.21260 0.07690  0.612 0.013
Doleromyrma.1             0.01349 0.02543 0.53070 0.00000 0.23080  0.630 0.434
Solenopsis.1              0.01196 0.02470 0.48430 0.20000 0.00000  0.645 0.056
Camponotus.consobrinus    0.01148 0.02157 0.53230 0.00000 0.24530  0.660 0.577
Iridomyrmex.purpureus     0.01099 0.02700 0.40730 0.00000 0.20030  0.675 0.914
Papyrius.2                0.01076 0.02258 0.47680 0.06670 0.15380  0.689 0.469
Machomyrma.4              0.01054 0.02600 0.40530 0.00000 0.15380  0.703 0.006
Prolasius.6               0.01052 0.02193 0.47970 0.06670 0.16840  0.717 0.341
Ochetellus.1              0.01026 0.02493 0.41170 0.00000 0.16840  0.730 0.647
Colobostruma.1            0.01014 0.02103 0.48230 0.20000 0.00000  0.743 0.018
Crematogaster.2           0.00923 0.02469 0.37390 0.06670 0.07690  0.755 0.765
Camponotus.12             0.00800 0.01933 0.41400 0.00000 0.15380  0.766 0.704
Mesostruma.1              0.00789 0.02143 0.36830 0.06670 0.09150  0.776 0.610
Rhopalomastix.3           0.00789 0.02085 0.37840 0.15860 0.00000  0.786 0.081
Mayriella.2               0.00715 0.01883 0.37960 0.06670 0.07690  0.796 0.688
Myrmecia.1                0.00686 0.01811 0.37880 0.13330 0.00000  0.805 0.128
Myrmecia.3                0.00674 0.01786 0.37720 0.13330 0.00000  0.814 0.528
Heteroponera.1            0.00670 0.01775 0.37760 0.13330 0.00000  0.822 0.819
Prolasius.3               0.00648 0.01681 0.38550 0.06670 0.07690  0.831 0.410
Doleromyrma.3             0.00605 0.01574 0.38450 0.13330 0.00000  0.839 0.877
Rhopalomastix.2           0.00593 0.02266 0.26160 0.09430 0.00000  0.847 0.271
Prolasius.7               0.00505 0.01806 0.27960 0.00000 0.07690  0.853 0.039
Stigmacros.2              0.00493 0.01745 0.28270 0.00000 0.09150  0.860 0.050
Anonychromyrma.2          0.00472 0.01681 0.28090 0.00000 0.07690  0.866 0.040
Polyrachis.7              0.00444 0.01577 0.28180 0.00000 0.07690  0.872 0.057
Stigmacros.1              0.00444 0.01577 0.28180 0.00000 0.07690  0.877 0.337
Pheidole.1                0.00441 0.01566 0.28190 0.00000 0.07690  0.883 0.049
Paratrechina.6            0.00432 0.01644 0.26270 0.07930 0.00000  0.889 0.280
Prolasius.2               0.00432 0.01644 0.26270 0.07930 0.00000  0.895 0.322
Prolasius.1               0.00399 0.01514 0.26330 0.07930 0.00000  0.900 0.608
Strumigenys.2             0.00398 0.01520 0.26200 0.06670 0.00000  0.905 0.278
Camponotis.26             0.00375 0.01320 0.28380 0.00000 0.07690  0.910 0.058
Camponotus.1              0.00375 0.01320 0.28380 0.00000 0.07690  0.915 0.058
Epopostruma.1             0.00375 0.01320 0.28380 0.00000 0.07690  0.920 0.058
Polyrachis.3              0.00375 0.01320 0.28380 0.00000 0.07690  0.925 0.525
Plagiolepis.1             0.00363 0.01383 0.26270 0.06670 0.00000  0.929 0.280
Anonychromyrma.3          0.00356 0.01252 0.28420 0.00000 0.07690  0.934 0.060
Myrmecorhrynchus.1        0.00356 0.01252 0.28420 0.00000 0.07690  0.939 0.060
Paratrechina.5            0.00356 0.01252 0.28420 0.00000 0.07690  0.944 0.060
Iridomyrmex.4             0.00356 0.01251 0.28420 0.00000 0.07690  0.948 0.738
Oligomyrmex.2             0.00356 0.01251 0.28420 0.00000 0.07690  0.953 0.204
Prolasius.4               0.00356 0.01251 0.28420 0.00000 0.07690  0.957 0.067
Notoncus.3                0.00352 0.01340 0.26290 0.06670 0.00000  0.962 0.276
Polyrachis.2              0.00352 0.01340 0.26290 0.06670 0.00000  0.967 0.605
Probolomyrmex.1           0.00338 0.01283 0.26320 0.06670 0.00000  0.971 0.273
Tetramorium.5             0.00338 0.01283 0.26320 0.06670 0.00000  0.976 0.273
Papyrius.1                0.00335 0.01273 0.26330 0.06670 0.00000  0.980 0.470
Pheidole.2                0.00335 0.01273 0.26330 0.06670 0.00000  0.984 0.284
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Leptomyrmex.1             0.00333 0.01265 0.26330 0.06670 0.00000  0.989 0.248
Stigmacros.3              0.00296 0.01035 0.28550 0.00000 0.07690  0.993 0.757
Mayriella.1               0.00284 0.01076 0.26410 0.06670 0.00000  0.996 0.557
Strumigenys.1             0.00284 0.01076 0.26410 0.06670 0.00000  1.000 0.742
Colobostruma.2            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Crematogaster.3           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Disturbed.lost            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Doleromyrma.2             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Dolichonderus.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Froggattella.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.2             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.5             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.6             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.7             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.8             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Machomyrma.3              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Machomyrma.6              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Melophorus.1              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Melophorus.2              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Meranoplus.2              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.2                0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.4                0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Notoncus.4                0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Oligomyrmex.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pachychondyla.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pachycondyla.2            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.6.1              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.1              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.5              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.8              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.5           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Stigmacros.4              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Technomyrmex.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA

Pheidole.5                *
Tetramorium.3             ***
Anonychromyrma.1          .
Rhytidiponera..metallica.
Pheidole.7
Crematogaster.1           **
Monomorium.1
Tapinoma.1
Meranoplus.1
Paratrechina.2
Machomyrma.1
Notoncus.1
Pheidole.6
Paratrechina.4
Paratrechina.1
Pheidole.7.1
Tetramorium.4             **
Pristomyrmex.2
Pristomyrmex.1            *
Doleromyrma.1
Solenopsis.1              .
Camponotus.consobrinus
Iridomyrmex.purpureus
Papyrius.2
Machomyrma.4              **
Prolasius.6
Ochetellus.1
Colobostruma.1            *
Crematogaster.2
Camponotus.12
```

```
Mesostruma.1
Rhopalomastix.3           .
Mayriella.2
Myrmecia.1
Myrmecia.3
Heteroponera.1
Prolasius.3
Doleromyrma.3
Rhopalomastix.2
Prolasius.7               *
Stigmacros.2              *
Anonychromyrma.2          *
Polyrachis.7              .
Stigmacros.1
Pheidole.1                *
Paratrechina.6
Prolasius.2
Prolasius.1
Strumigenys.2
Camponotis.26             .
Camponotus.1              .
Epopostruma.1             .
Polyrachis.3
Plagiolepis.1
Anonychromyrma.3          .
Myrmecorhrynchus.1        .
Paratrechina.5            .
Iridomyrmex.4
Oligomyrmex.2
Prolasius.4               .
Notoncus.3
Polyrachis.2
Probolomyrmex.1
Tetramorium.5
Papyrius.1
Pheidole.2
Leptomyrmex.1
Stigmacros.3
Mayriella.1
Strumigenys.1
Colobostruma.2
Crematogaster.3
Disturbed.lost
Doleromyrma.2
Dolichonderus.1
Froggattella.1
Iridomyrmex.2
Iridomyrmex.5
Iridomyrmex.6
Iridomyrmex.7
Iridomyrmex.8
Machomyrma.3
Machomyrma.6
Melophorus.1
Melophorus.2
Meranoplus.2
Myrmecia.2
Myrmecia.4
Notoncus.4
Oligomyrmex.1
Pachychondyla.1
Pachycondyla.2
Pheidole.6.1
Polyrachis.1
Polyrachis.5
```

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Polyrachis.8
Rhopalomastix.1
Rhopalomastix.5
Stigmacros.4
Technomyrmex.1
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: CCIF_CPW

                          average      sd   ratio     ava     avb cumsum     p
Pheidole.5                0.03206 0.03256 0.98460 1.22750 0.92190  0.047 0.940
Tapinoma.1                0.03099 0.02800 1.10670 0.65120 0.73330  0.093 0.494
Meranoplus.1              0.02893 0.02737 1.05720 0.33620 0.65660  0.136 0.326
Iridomyrmex.purpureus     0.02750 0.02985 0.92150 0.26180 0.54280  0.176 0.029
Paratrechina.1            0.02696 0.02684 1.00460 0.25000 0.57810  0.216 0.170
Notoncus.1                0.02613 0.02863 0.91270 0.37000 0.42060  0.255 0.615
Machomyrma.1              0.02456 0.02646 0.92830 0.33620 0.43750  0.291 0.125
Pheidole.7                0.02400 0.02762 0.86890 0.38970 0.34410  0.327 0.825
Mesostruma.1              0.01697 0.02569 0.66080 0.00000 0.39440  0.352 0.027
Monomorium.1              0.01693 0.02906 0.58250 0.29450 0.16760  0.377 0.886
Iridomyrmex.5             0.01657 0.02626 0.63080 0.00000 0.34800  0.401 0.002
Ochetellus.1              0.01623 0.02231 0.72760 0.12500 0.32430  0.425 0.109
Camponotus.consobrinus    0.01607 0.02473 0.64990 0.12500 0.25000  0.449 0.153
Iridomyrmex.2             0.01595 0.02432 0.65580 0.12500 0.27370  0.473 0.091
Rhytidiponera..metallica. 0.01589 0.02204 0.72090 1.30250 1.45860  0.496 1.000
Anonychromyrma.1          0.01553 0.03085 0.50350 0.25380 0.08230  0.519 0.996
Camponotus.12             0.01540 0.02177 0.70720 0.19930 0.25000  0.542 0.049
Doleromyrma.1             0.01473 0.02288 0.64370 0.18750 0.19930  0.564 0.257
Paratrechina.2            0.01424 0.02191 0.64980 0.12500 0.25000  0.585 0.909
Pheidole.6                0.01356 0.02101 0.64540 0.19930 0.18750  0.605 0.382
Papyrius.2                0.01304 0.02789 0.46770 0.00000 0.21120  0.624 0.262
Prolasius.6               0.01288 0.02309 0.55760 0.00000 0.31540  0.643 0.130
Mayriella.2               0.01279 0.02114 0.60490 0.06250 0.25000  0.662 0.161
Iridomyrmex.7             0.01210 0.02174 0.55680 0.26180 0.00000  0.680 0.021
Crematogaster.1           0.01193 0.02094 0.56950 0.00000 0.25000  0.697 0.923
Pristomyrmex.2            0.01182 0.02325 0.50820 0.07430 0.22520  0.715 0.724
Heteroponera.1            0.01130 0.02161 0.52300 0.18750 0.07430  0.731 0.449
Stigmacros.3              0.01053 0.01998 0.52680 0.06250 0.21910  0.747 0.036
Pheidole.7.1              0.00985 0.02127 0.46320 0.00000 0.19930  0.761 0.584
Polyrachis.5              0.00969 0.02078 0.46630 0.18750 0.00000  0.776 0.076
Rhopalomastix.1           0.00964 0.02216 0.43480 0.14480 0.07430  0.790 0.164
Machomyrma.3              0.00950 0.02072 0.45850 0.18750 0.00000  0.804 0.299
Crematogaster.3           0.00837 0.02340 0.35770 0.16760 0.00000  0.817 0.182
Paratrechina.4            0.00795 0.01827 0.43500 0.14480 0.06250  0.828 0.863
Iridomyrmex.4             0.00730 0.02037 0.35830 0.06250 0.09350  0.839 0.337
Polyrachis.3              0.00729 0.01635 0.44580 0.06250 0.12500  0.850 0.199
Polyrachis.2              0.00676 0.01508 0.44830 0.06250 0.12500  0.860 0.202
Notoncus.4                0.00655 0.01775 0.36910 0.13680 0.00000  0.870 0.194
Polyrachis.8              0.00626 0.01674 0.37380 0.00000 0.12500  0.879 0.025
Pachycondyla.2            0.00536 0.01456 0.36830 0.00000 0.12500  0.887 0.498
Doleromyrma.3             0.00513 0.01372 0.37380 0.00000 0.12500  0.894 0.934
Melophorus.2              0.00481 0.01332 0.36160 0.06250 0.07430  0.901 0.169
Papyrius.1                0.00474 0.01322 0.35850 0.06250 0.06250  0.908 0.389
Prolasius.1               0.00460 0.01285 0.35820 0.06250 0.06250  0.915 0.558
Myrmecia.2                0.00338 0.01323 0.25520 0.00000 0.06250  0.920 0.197
Doleromyrma.2             0.00336 0.01335 0.25140 0.06250 0.00000  0.925 0.424
Polyrachis.1              0.00327 0.01298 0.25170 0.06250 0.00000  0.930 0.433
Rhopalomastix.2           0.00322 0.01277 0.25190 0.06250 0.00000  0.935 0.645
Mayriella.1               0.00314 0.01228 0.25550 0.00000 0.06250  0.939 0.263
Oligomyrmex.1             0.00314 0.01228 0.25550 0.00000 0.06250  0.944 0.163
Melophorus.1              0.00310 0.01228 0.25230 0.06250 0.00000  0.949 0.428
Stigmacros.4              0.00296 0.01157 0.25570 0.00000 0.06250  0.953 0.516
Iridomyrmex.6             0.00292 0.01143 0.25580 0.00000 0.06250  0.957 0.188
Froggattella.1            0.00283 0.01118 0.25310 0.06250 0.00000  0.961 0.408
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Strumigenys.1            0.00283 0.01118 0.25310 0.06250 0.00000  0.966 0.665
Meranoplus.2             0.00275 0.01072 0.25600 0.00000 0.06250  0.970 0.204
Myrmecia.3               0.00275 0.01072 0.25600 0.00000 0.06250  0.974 0.884
Colobostruma.2           0.00263 0.01036 0.25370 0.06250 0.00000  0.978 0.427
Machomyrma.6             0.00262 0.01033 0.25370 0.06250 0.00000  0.981 0.421
Pheidole.6.1             0.00238 0.00929 0.25640 0.00000 0.06250  0.985 0.187
Iridomyrmex.8            0.00204 0.00794 0.25670 0.00000 0.06250  0.988 0.193
Myrmecia.1               0.00204 0.00794 0.25670 0.00000 0.06250  0.991 0.760
Myrmecia.4               0.00204 0.00794 0.25670 0.00000 0.06250  0.994 0.193
Rhopalomastix.5          0.00204 0.00794 0.25670 0.00000 0.06250  0.997 0.193
Tetramorium.4            0.00204 0.00794 0.25670 0.00000 0.06250  1.000 0.936
Anonychromyrma.2         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Anonychromyrma.3         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Camponotis.26            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Camponotus.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Colobostruma.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Crematogaster.2          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Disturbed.lost           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Dolichonderus.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Epopostruma.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Leptomyrmex.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Machomyrma.4             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecorhrynchus.1       0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Notoncus.3               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Oligomyrmex.2            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pachychondyla.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Paratrechina.5           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Paratrechina.6           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.1               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.2               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Plagiolepis.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.7             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pristomyrmex.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Probolomyrmex.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.2              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.3              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.4              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.7              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.3          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Solenopsis.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Stigmacros.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Stigmacros.2             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Strumigenys.2            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Technomyrmex.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Tetramorium.3            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Tetramorium.5            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA

Pheidole.5
Tapinoma.1
Meranoplus.1
Iridomyrmex.purpureus    *
Paratrechina.1
Notoncus.1
Machomyrma.1
Pheidole.7
Mesostruma.1             *
Monomorium.1
Iridomyrmex.5            **
Ochetellus.1
Camponotus.consobrinus
Iridomyrmex.2            .
Rhytidiponera..metallica.
Anonychromyrma.1
Camponotus.12            *
Doleromyrma.1
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Paratrechina.2
Pheidole.6
Papyrius.2
Prolasius.6
Mayriella.2
Iridomyrmex.7             *
Crematogaster.1
Pristomyrmex.2
Heteroponera.1
Stigmacros.3              *
Pheidole.7.1
Polyrachis.5              .
Rhopalomastix.1
Machomyrma.3
Crematogaster.3
Paratrechina.4
Iridomyrmex.4
Polyrachis.3
Polyrachis.2
Notoncus.4
Polyrachis.8              *
Pachycondyla.2
Doleromyrma.3
Melophorus.2
Papyrius.1
Prolasius.1
Myrmecia.2
Doleromyrma.2
Polyrachis.1
Rhopalomastix.2
Mayriella.1
Oligomyrmex.1
Melophorus.1
Stigmacros.4
Iridomyrmex.6
Froggattella.1
Strumigenys.1
Meranoplus.2
Myrmecia.3
Colobostruma.2
Machomyrma.6
Pheidole.6.1
Iridomyrmex.8
Myrmecia.1
Myrmecia.4
Rhopalomastix.5
Tetramorium.4
Anonychromyrma.2
Anonychromyrma.3
Camponotis.26
Camponotus.1
Colobostruma.1
Crematogaster.2
Disturbed.lost
Dolichonderus.1
Epopostruma.1
Leptomyrmex.1
Machomyrma.4
Myrmecorhrynchus.1
Notoncus.3
Oligomyrmex.2
Pachychondyla.1
Paratrechina.5
Paratrechina.6
Pheidole.1
```

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Pheidole.2
Plagiolepis.1
Polyrachis.7
Pristomyrmex.1
Probolomyrmex.1
Prolasius.2
Prolasius.3
Prolasius.4
Prolasius.7
Rhopalomastix.3
Solenopsis.1
Stigmacros.1
Stigmacros.2
Strumigenys.2
Technomyrmex.1
Tetramorium.3
Tetramorium.5
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: CCIF_SRW

                          average      sd   ratio     ava     avb cumsum     p
Rhytidiponera..metallica. 0.04257 0.03775 1.12770 1.30250 0.79590  0.057 0.020
Pheidole.5                0.04182 0.03620 1.15540 1.22750 0.75730  0.113 0.152
Tapinoma.1                0.03669 0.03424 1.07160 0.65120 0.18750  0.163 0.018
Notoncus.1                0.03334 0.03491 0.95500 0.37000 0.48480  0.207 0.034
Anonychromyrma.1          0.03050 0.04018 0.75920 0.25380 0.35530  0.248 0.409
Pheidole.7                0.02855 0.03471 0.82260 0.38970 0.26180  0.287 0.434
Meranoplus.1              0.02722 0.03452 0.78840 0.33620 0.24430  0.323 0.507
Crematogaster.2           0.02652 0.03561 0.74470 0.00000 0.43640  0.359 0.003
Monomorium.1              0.02564 0.03936 0.65160 0.29450 0.27240  0.393 0.354
Paratrechina.1            0.02503 0.03087 0.81080 0.25000 0.34410  0.427 0.400
Doleromyrma.3             0.02279 0.04130 0.55180 0.00000 0.33670  0.458 0.024
Machomyrma.1              0.02166 0.03247 0.66710 0.33620 0.06250  0.487 0.348
Paratrechina.2            0.02106 0.02861 0.73610 0.12500 0.33230  0.515 0.321
Iridomyrmex.purpureus     0.01723 0.02873 0.59990 0.26180 0.06250  0.538 0.617
Heteroponera.1            0.01655 0.02894 0.57170 0.18750 0.12500  0.560 0.044
Machomyrma.3              0.01645 0.02882 0.57080 0.18750 0.12500  0.583 0.005
Tetramorium.3             0.01577 0.02844 0.55440 0.00000 0.29710  0.604 0.933
Iridomyrmex.7             0.01509 0.02700 0.55880 0.26180 0.00000  0.624 0.001
Paratrechina.4            0.01444 0.02783 0.51880 0.14480 0.12500  0.643 0.352
Doleromyrma.1             0.01419 0.02703 0.52470 0.18750 0.06250  0.662 0.358
Iridomyrmex.2             0.01410 0.02789 0.50560 0.12500 0.12500  0.681 0.235
Pristomyrmex.2            0.01367 0.02639 0.51790 0.07430 0.21340  0.700 0.573
Polyrachis.5              0.01242 0.02652 0.46830 0.18750 0.00000  0.716 0.001
Rhopalomastix.1           0.01228 0.02825 0.43460 0.14480 0.06250  0.733 0.035
Pheidole.6                0.01166 0.02200 0.52990 0.19930 0.06250  0.749 0.559
Crematogaster.3           0.01062 0.02951 0.35990 0.16760 0.00000  0.763 0.001
Camponotus.consobrinus    0.00998 0.02307 0.43250 0.12500 0.06250  0.776 0.736
Camponotus.12             0.00976 0.02087 0.46790 0.19930 0.00000  0.789 0.534
Ochetellus.1              0.00876 0.01988 0.44060 0.12500 0.06250  0.801 0.771
Notoncus.4                0.00824 0.02220 0.37100 0.13680 0.00000  0.812 0.005
Technomyrmex.1            0.00806 0.02172 0.37120 0.00000 0.12500  0.823 0.030
Myrmecia.3                0.00804 0.02185 0.36780 0.00000 0.12500  0.834 0.362
Pachycondyla.2            0.00793 0.02127 0.37260 0.00000 0.13680  0.844 0.202
Prolasius.3               0.00733 0.01982 0.36990 0.00000 0.13680  0.854 0.232
Iridomyrmex.4             0.00717 0.02019 0.35530 0.06250 0.06250  0.864 0.361
Dolichonderus.1           0.00707 0.01899 0.37250 0.00000 0.12500  0.873 0.045
Stigmacros.1              0.00679 0.01819 0.37320 0.00000 0.12500  0.882 0.191
Strumigenys.1             0.00657 0.01830 0.35900 0.06250 0.06250  0.891 0.103
Prolasius.1               0.00650 0.01816 0.35820 0.06250 0.06250  0.900 0.312
Mayriella.2               0.00630 0.01755 0.35920 0.06250 0.06250  0.909 0.810
Doleromyrma.2             0.00434 0.01721 0.25250 0.06250 0.00000  0.914 0.007
Polyrachis.1              0.00420 0.01660 0.25280 0.06250 0.00000  0.920 0.010
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Rhopalomastix.2          0.00412 0.01626 0.25300 0.06250 0.00000  0.926 0.415
Melophorus.1             0.00392 0.01548 0.25350 0.06250 0.00000  0.931 0.002
Crematogaster.1          0.00390 0.01533 0.25440 0.00000 0.06250  0.936 1.000
Froggattella.1           0.00351 0.01380 0.25430 0.06250 0.00000  0.941 0.006
Papyrius.2               0.00346 0.01356 0.25510 0.00000 0.06250  0.945 0.941
Solenopsis.1             0.00344 0.01348 0.25510 0.00000 0.06250  0.950 0.766
Pachychondyla.1          0.00339 0.01327 0.25520 0.00000 0.06250  0.955 0.207
Polyrachis.2             0.00328 0.01289 0.25480 0.06250 0.00000  0.959 0.630
Colobostruma.2           0.00320 0.01257 0.25490 0.06250 0.00000  0.963 0.009
Pheidole.7.1             0.00320 0.01253 0.25540 0.00000 0.06250  0.968 0.973
Prolasius.2              0.00320 0.01253 0.25540 0.00000 0.06250  0.972 0.547
Stigmacros.4             0.00320 0.01253 0.25540 0.00000 0.06250  0.976 0.283
Machomyrma.6             0.00320 0.01253 0.25490 0.06250 0.00000  0.981 0.006
Melophorus.2             0.00320 0.01253 0.25490 0.06250 0.00000  0.985 0.164
Polyrachis.3             0.00320 0.01253 0.25490 0.06250 0.00000  0.989 0.560
Oligomyrmex.2            0.00278 0.01087 0.25590 0.00000 0.06250  0.993 0.460
Papyrius.1               0.00267 0.01044 0.25580 0.06250 0.00000  0.996 0.636
Stigmacros.3             0.00267 0.01044 0.25580 0.06250 0.00000  1.000 0.781
Anonychromyrma.2         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Anonychromyrma.3         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Camponotis.26            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Camponotus.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Colobostruma.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Disturbed.lost           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Epopostruma.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.5            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.6            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.8            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Leptomyrmex.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Machomyrma.4             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Mayriella.1              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Meranoplus.2             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Mesostruma.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.1               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.2               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.4               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecorhrynchus.1       0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Notoncus.3               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Oligomyrmex.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Paratrechina.5           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Paratrechina.6           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.1               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.2               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.6.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Plagiolepis.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.7             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.8             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pristomyrmex.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Probolomyrmex.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.4              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.6              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.7              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.3          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.5          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Stigmacros.2             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Strumigenys.2            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Tetramorium.4            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Tetramorium.5            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA

Rhytidiponera..metallica. *
Pheidole.5
Tapinoma.1               *
Notoncus.1               *
Anonychromyrma.1
Pheidole.7
```

```
Meranoplus.1
Crematogaster.2           **
Monomorium.1
Paratrechina.1
Doleromyrma.3             *
Machomyrma.1
Paratrechina.2
Iridomyrmex.purpureus
Heteroponera.1            *
Machomyrma.3              **
Tetramorium.3
Iridomyrmex.7             ***
Paratrechina.4
Doleromyrma.1
Iridomyrmex.2
Pristomyrmex.2
Polyrachis.5              ***
Rhopalomastix.1           *
Pheidole.6
Crematogaster.3           ***
Camponotus.consobrinus
Camponotus.12
Ochetellus.1
Notoncus.4                **
Technomyrmex.1            *
Myrmecia.3
Pachycondyla.2
Prolasius.3
Iridomyrmex.4
Dolichonderus.1           *
Stigmacros.1
Strumigenys.1
Prolasius.1
Mayriella.2
Doleromyrma.2             **
Polyrachis.1              **
Rhopalomastix.2
Melophorus.1              **
Crematogaster.1
Froggattella.1            **
Papyrius.2
Solenopsis.1
Pachychondyla.1
Polyrachis.2
Colobostruma.2            **
Pheidole.7.1
Prolasius.2
Stigmacros.4
Machomyrma.6              **
Melophorus.2
Polyrachis.3
Oligomyrmex.2
Papyrius.1
Stigmacros.3
Anonychromyrma.2
Anonychromyrma.3
Camponotis.26
Camponotus.1
Colobostruma.1
Disturbed.lost
Epopostruma.1
Iridomyrmex.5
Iridomyrmex.6
Iridomyrmex.8
Leptomyrmex.1
```

```
Machomyrma.4
Mayriella.1
Meranoplus.2
Mesostruma.1
Myrmecia.1
Myrmecia.2
Myrmecia.4
Myrmecorhrynchus.1
Notoncus.3
Oligomyrmex.1
Paratrechina.5
Paratrechina.6
Pheidole.1
Pheidole.2
Pheidole.6.1
Plagiolepis.1
Polyrachis.7
Polyrachis.8
Pristomyrmex.1
Probolomyrmex.1
Prolasius.4
Prolasius.6
Prolasius.7
Rhopalomastix.3
Rhopalomastix.5
Stigmacros.2
Strumigenys.2
Tetramorium.4
Tetramorium.5
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: CCIF_SSTF

                          average      sd   ratio     ava     avb cumsum     p
Tapinoma.1                0.03262 0.03060 1.06610 0.65120 0.38460  0.048 0.302
Anonychromyrma.1          0.03136 0.03725 0.84210 0.25380 0.49420  0.094 0.334
Crematogaster.1           0.02882 0.03195 0.90180 0.00000 0.56960  0.136 0.004
Pheidole.5                0.02692 0.03286 0.81920 1.22750 1.08140  0.175 0.993
Machomyrma.1              0.02656 0.02988 0.88910 0.33620 0.41370  0.214 0.048
Pheidole.7                0.02651 0.03231 0.82060 0.38970 0.24530  0.253 0.643
Notoncus.1                0.02578 0.03264 0.78980 0.37000 0.24530  0.290 0.632
Rhytidiponera..metallica. 0.02298 0.03038 0.75630 1.30250 1.22640  0.324 0.922
Meranoplus.1              0.02210 0.02966 0.74510 0.33620 0.15380  0.356 0.874
Iridomyrmex.purpureus     0.02041 0.03009 0.67830 0.26180 0.20030  0.386 0.401
Pheidole.6                0.02040 0.02656 0.76820 0.19930 0.32220  0.416 0.020
Monomorium.1              0.02023 0.03295 0.61410 0.29450 0.20030  0.446 0.675
Paratrechina.2            0.01989 0.02774 0.71710 0.12500 0.32220  0.474 0.474
Doleromyrma.1             0.01899 0.02775 0.68420 0.18750 0.23080  0.502 0.068
Paratrechina.1            0.01753 0.02640 0.66420 0.25000 0.15380  0.528 0.933
Camponotus.consobrinus    0.01541 0.02442 0.63080 0.12500 0.24530  0.550 0.242
Tetramorium.3             0.01462 0.02929 0.49920 0.00000 0.25510  0.572 0.929
Camponotus.12             0.01447 0.02343 0.61770 0.19930 0.15380  0.593 0.114
Ochetellus.1              0.01412 0.02598 0.54340 0.12500 0.16840  0.614 0.286
Iridomyrmex.7             0.01378 0.02463 0.55940 0.26180 0.00000  0.634 0.007
Polyrachis.5              0.01120 0.02387 0.46900 0.18750 0.00000  0.650 0.040
Heteroponera.1            0.01100 0.02391 0.45990 0.18750 0.00000  0.666 0.471
Machomyrma.3              0.01099 0.02394 0.45900 0.18750 0.00000  0.682 0.162
Paratrechina.4            0.01052 0.02267 0.46410 0.14480 0.09150  0.698 0.691
Machomyrma.4              0.01022 0.02464 0.41490 0.00000 0.15380  0.713 0.010
Pristomyrmex.2            0.00983 0.02038 0.48230 0.07430 0.15380  0.727 0.817
Crematogaster.3           0.00962 0.02673 0.35990 0.16760 0.00000  0.741 0.118
Rhopalomastix.1           0.00884 0.02426 0.36460 0.14480 0.00000  0.754 0.239
Papyrius.2                0.00861 0.02074 0.41500 0.00000 0.15380  0.767 0.637
Prolasius.6               0.00777 0.01909 0.40710 0.00000 0.16840  0.778 0.610
```

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Notoncus.4            0.00749 0.02019 0.37110 0.13680 0.00000  0.789 0.127
Iridomyrmex.4         0.00682 0.01819 0.37510 0.06250 0.07690  0.799 0.444
Iridomyrmex.2         0.00664 0.01802 0.36860 0.12500 0.00000  0.809 0.858
Polyrachis.3          0.00624 0.01635 0.38150 0.06250 0.07690  0.818 0.251
Mayriella.2           0.00607 0.01591 0.38170 0.06250 0.07690  0.827 0.839
Crematogaster.2       0.00570 0.02016 0.28280 0.00000 0.07690  0.835 0.898
Mesostruma.1          0.00549 0.01929 0.28450 0.00000 0.09150  0.843 0.810
Stigmacros.3          0.00513 0.01339 0.38320 0.06250 0.07690  0.850 0.556
Prolasius.7           0.00492 0.01732 0.28400 0.00000 0.07690  0.858 0.091
Stigmacros.2          0.00485 0.01699 0.28520 0.00000 0.09150  0.865 0.082
Anonychromyrma.2      0.00461 0.01622 0.28450 0.00000 0.07690  0.871 0.085
Polyrachis.7          0.00435 0.01529 0.28490 0.00000 0.07690  0.878 0.075
Stigmacros.1          0.00435 0.01529 0.28490 0.00000 0.07690  0.884 0.369
Pheidole.1            0.00433 0.01518 0.28490 0.00000 0.07690  0.890 0.100
Pristomyrmex.1        0.00407 0.01429 0.28520 0.00000 0.07690  0.896 0.625
Doleromyrma.2         0.00390 0.01542 0.25280 0.06250 0.00000  0.902 0.260
Polyrachis.1          0.00378 0.01493 0.25310 0.06250 0.00000  0.908 0.274
Rhopalomastix.2       0.00371 0.01466 0.25320 0.06250 0.00000  0.913 0.565
Camponotis.26         0.00369 0.01292 0.28570 0.00000 0.07690  0.918 0.087
Camponotus.1          0.00369 0.01292 0.28570 0.00000 0.07690  0.924 0.087
Epopostruma.1         0.00369 0.01292 0.28570 0.00000 0.07690  0.929 0.087
Melophorus.1          0.00356 0.01402 0.25360 0.06250 0.00000  0.934 0.286
Anonychromyrma.3      0.00351 0.01228 0.28590 0.00000 0.07690  0.940 0.081
Myrmecorhrynchus.1    0.00351 0.01228 0.28590 0.00000 0.07690  0.945 0.081
Paratrechina.5        0.00351 0.01228 0.28590 0.00000 0.07690  0.950 0.081
Prolasius.3           0.00351 0.01228 0.28590 0.00000 0.07690  0.955 0.705
Oligomyrmex.2         0.00351 0.01227 0.28590 0.00000 0.07690  0.960 0.214
Prolasius.4           0.00351 0.01227 0.28590 0.00000 0.07690  0.965 0.076
Froggattella.1        0.00321 0.01264 0.25430 0.06250 0.00000  0.970 0.293
Strumigenys.1         0.00321 0.01264 0.25430 0.06250 0.00000  0.975 0.656
Polyrachis.2          0.00302 0.01187 0.25460 0.06250 0.00000  0.979 0.672
Colobostruma.2        0.00296 0.01160 0.25470 0.06250 0.00000  0.983 0.281
Prolasius.1           0.00296 0.01160 0.25470 0.06250 0.00000  0.988 0.763
Machomyrma.6          0.00295 0.01157 0.25470 0.06250 0.00000  0.992 0.278
Melophorus.2          0.00295 0.01157 0.25470 0.06250 0.00000  0.996 0.408
Papyrius.1            0.00250 0.00977 0.25550 0.06250 0.00000  1.000 0.733
Colobostruma.1        0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Disturbed.lost        0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Doleromyrma.3         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Dolichonderus.1       0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.5         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.6         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.8         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Leptomyrmex.1         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Mayriella.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Meranoplus.2          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.2            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.3            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.4            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Notoncus.3            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Oligomyrmex.1         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pachychondyla.1       0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pachycondyla.2        0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Paratrechina.6        0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.2            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.6.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.7.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Plagiolepis.1         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.8          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Probolomyrmex.1       0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.2           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.3       0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.5       0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Solenopsis.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
```

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Stigmacros.4              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Strumigenys.2             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Technomyrmex.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Tetramorium.4             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Tetramorium.5             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA

Tapinoma.1
Anonychromyrma.1
Crematogaster.1           **
Pheidole.5
Machomyrma.1              *
Pheidole.7
Notoncus.1
Rhytidiponera..metallica.
Meranoplus.1
Iridomyrmex.purpureus
Pheidole.6                *
Monomorium.1
Paratrechina.2
Doleromyrma.1             .
Paratrechina.1
Camponotus.consobrinus
Tetramorium.3
Camponotus.12
Ochetellus.1
Iridomyrmex.7             **
Polyrachis.5              *
Heteroponera.1
Machomyrma.3
Paratrechina.4
Machomyrma.4              **
Pristomyrmex.2
Crematogaster.3
Rhopalomastix.1
Papyrius.2
Prolasius.6
Notoncus.4
Iridomyrmex.4
Iridomyrmex.2
Polyrachis.3
Mayriella.2
Crematogaster.2
Mesostruma.1
Stigmacros.3
Prolasius.7               .
Stigmacros.2              .
Anonychromyrma.2          .
Polyrachis.7              .
Stigmacros.1
Pheidole.1                .
Pristomyrmex.1
Doleromyrma.2
Polyrachis.1
Rhopalomastix.2
Camponotis.26             .
Camponotus.1              .
Epopostruma.1             .
Melophorus.1
Anonychromyrma.3          .
Myrmecorhrynchus.1        .
Paratrechina.5            .
Prolasius.3
Oligomyrmex.2
Prolasius.4               .
Froggattella.1
```

```
Strumigenys.1
Polyrachis.2
Colobostruma.2
Prolasius.1
Machomyrma.6
Melophorus.2
Papyrius.1
Colobostruma.1
Disturbed.lost
Doleromyrma.3
Dolichonderus.1
Iridomyrmex.5
Iridomyrmex.6
Iridomyrmex.8
Leptomyrmex.1
Mayriella.1
Meranoplus.2
Myrmecia.1
Myrmecia.2
Myrmecia.3
Myrmecia.4
Notoncus.3
Oligomyrmex.1
Pachychondyla.1
Pachycondyla.2
Paratrechina.6
Pheidole.2
Pheidole.6.1
Pheidole.7.1
Plagiolepis.1
Polyrachis.8
Probolomyrmex.1
Prolasius.2
Rhopalomastix.3
Rhopalomastix.5
Solenopsis.1
Stigmacros.4
Strumigenys.2
Technomyrmex.1
Tetramorium.4
Tetramorium.5
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: CPW_SRW

                          average      sd   ratio     ava     avb cumsum     p
Rhytidiponera..metallica. 0.03822 0.03415 1.11900 1.45860 0.79590  0.050 0.082
Pheidole.5                0.03590 0.03273 1.09680 0.92190 0.75730  0.097 0.696
Tapinoma.1                0.03476 0.03181 1.09270 0.73330 0.18750  0.142 0.068
Meranoplus.1              0.03390 0.03020 1.12240 0.65660 0.24430  0.186 0.032
Paratrechina.1            0.03007 0.02943 1.02200 0.57810 0.34410  0.225 0.027
Notoncus.1                0.02899 0.02993 0.96880 0.42060 0.48480  0.263 0.317
Iridomyrmex.purpureus     0.02813 0.03357 0.83820 0.54280 0.06250  0.300 0.015
Machomyrma.1              0.02380 0.02847 0.83590 0.43750 0.06250  0.331 0.162
Crematogaster.2           0.02281 0.03108 0.73380 0.00000 0.43640  0.361 0.024
Doleromyrma.3             0.02183 0.03467 0.62960 0.12500 0.33670  0.389 0.040
Pheidole.7                0.02160 0.02632 0.82050 0.34410 0.26180  0.417 0.919
Paratrechina.2            0.02071 0.02575 0.80440 0.25000 0.33230  0.444 0.376
Anonychromyrma.1          0.01960 0.02853 0.68720 0.08230 0.35530  0.470 0.978
Mesostruma.1              0.01824 0.02777 0.65690 0.39440 0.00000  0.494 0.005
Iridomyrmex.5             0.01798 0.02888 0.62260 0.34800 0.00000  0.517 0.001
Iridomyrmex.2             0.01790 0.02762 0.64800 0.27370 0.12500  0.540 0.034
Pristomyrmex.2            0.01765 0.02843 0.62100 0.22520 0.21340  0.563 0.247
Monomorium.1              0.01651 0.02851 0.57900 0.16760 0.27240  0.585 0.888
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Papyrius.2              0.01607 0.03135 0.51270 0.21120 0.06250  0.606 0.081
Camponotus.consobrinus  0.01606 0.02662 0.60330 0.25000 0.06250  0.627 0.185
Ochetellus.1            0.01592 0.02311 0.68860 0.32430 0.06250  0.648 0.116
Crematogaster.1         0.01460 0.02391 0.61060 0.25000 0.06250  0.667 0.795
Mayriella.2             0.01391 0.02317 0.60040 0.25000 0.06250  0.685 0.088
Tetramorium.3           0.01377 0.02512 0.54840 0.00000 0.29710  0.703 0.956
Prolasius.6             0.01375 0.02477 0.55500 0.31540 0.00000  0.721 0.090
Pheidole.7.1            0.01245 0.02412 0.51630 0.19930 0.06250  0.737 0.377
Camponotus.12           0.01152 0.02073 0.55600 0.25000 0.00000  0.752 0.339
Pachycondyla.2          0.01123 0.02188 0.51330 0.12500 0.13680  0.767 0.035
Doleromyrma.1           0.01093 0.02069 0.52810 0.19930 0.06250  0.781 0.670
Pheidole.6              0.01029 0.01974 0.52150 0.18750 0.06250  0.794 0.736
Stigmacros.3            0.00964 0.02077 0.46410 0.21910 0.00000  0.807 0.091
Myrmecia.3              0.00905 0.02061 0.43880 0.06250 0.12500  0.819 0.288
Paratrechina.4          0.00870 0.02024 0.42990 0.06250 0.12500  0.830 0.825
Heteroponera.1          0.00839 0.01875 0.44750 0.07430 0.12500  0.841 0.714
Iridomyrmex.4           0.00740 0.02104 0.35190 0.09350 0.06250  0.851 0.306
Technomyrmex.1          0.00689 0.01879 0.36650 0.00000 0.12500  0.860 0.206
Polyrachis.8            0.00678 0.01822 0.37230 0.12500 0.00000  0.868 0.001
Prolasius.3             0.00640 0.01748 0.36640 0.00000 0.13680  0.877 0.412
Dolichonderus.1         0.00614 0.01666 0.36850 0.00000 0.12500  0.885 0.150
Machomyrma.3            0.00604 0.01635 0.36910 0.00000 0.12500  0.893 0.652
Stigmacros.1            0.00592 0.01603 0.36940 0.00000 0.12500  0.900 0.369
Stigmacros.4            0.00566 0.01584 0.35730 0.06250 0.06250  0.908 0.136
Polyrachis.3            0.00561 0.01525 0.36800 0.12500 0.00000  0.915 0.330
Rhopalomastix.1         0.00545 0.01512 0.36070 0.07430 0.06250  0.922 0.589
Prolasius.1             0.00525 0.01488 0.35280 0.06250 0.06250  0.929 0.517
Polyrachis.2            0.00488 0.01303 0.37440 0.12500 0.00000  0.935 0.489
Myrmecia.2              0.00369 0.01451 0.25400 0.06250 0.00000  0.940 0.012
Mayriella.1             0.00340 0.01337 0.25450 0.06250 0.00000  0.945 0.202
Oligomyrmex.1           0.00340 0.01337 0.25450 0.06250 0.00000  0.949 0.002
Iridomyrmex.6           0.00315 0.01236 0.25500 0.06250 0.00000  0.953 0.005
Strumigenys.1           0.00301 0.01192 0.25260 0.00000 0.06250  0.957 0.664
Solenopsis.1            0.00299 0.01185 0.25260 0.00000 0.06250  0.961 0.849
Pachychondyla.1         0.00295 0.01169 0.25270 0.00000 0.06250  0.965 0.412
Meranoplus.2            0.00294 0.01153 0.25530 0.06250 0.00000  0.969 0.009
Papyrius.1              0.00294 0.01153 0.25530 0.06250 0.00000  0.973 0.537
Prolasius.2             0.00281 0.01110 0.25320 0.00000 0.06250  0.976 0.649
Melophorus.2            0.00255 0.00994 0.25650 0.07430 0.00000  0.980 0.405
Pheidole.6.1            0.00253 0.00988 0.25600 0.06250 0.00000  0.983 0.006
Oligomyrmex.2           0.00248 0.00975 0.25410 0.00000 0.06250  0.986 0.657
Iridomyrmex.8           0.00214 0.00836 0.25650 0.06250 0.00000  0.989 0.006
Myrmecia.1              0.00214 0.00836 0.25650 0.06250 0.00000  0.992 0.700
Myrmecia.4              0.00214 0.00836 0.25650 0.06250 0.00000  0.994 0.006
Rhopalomastix.5         0.00214 0.00836 0.25650 0.06250 0.00000  0.997 0.006
Tetramorium.4           0.00214 0.00836 0.25650 0.06250 0.00000  1.000 0.901
Anonychromyrma.2        0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Anonychromyrma.3        0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Camponotis.26           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Camponotus.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Colobostruma.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Colobostruma.2          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Crematogaster.3         0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Disturbed.lost          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Doleromyrma.2           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Epopostruma.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Froggattella.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.7           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Leptomyrmex.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Machomyrma.4            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Machomyrma.6            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Melophorus.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecorhrynchus.1      0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Notoncus.3              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Notoncus.4              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
```

```
Paratrechina.5           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Paratrechina.6           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.1               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.2               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Plagiolepis.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.5             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.7             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pristomyrmex.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Probolomyrmex.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.4              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Prolasius.7              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.2          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.3          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Stigmacros.2             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Strumigenys.2            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Tetramorium.5            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA

Rhytidiponera..metallica. .
Pheidole.5
Tapinoma.1               .
Meranoplus.1             *
Paratrechina.1           *
Notoncus.1
Iridomyrmex.purpureus    *
Machomyrma.1
Crematogaster.2          *
Doleromyrma.3            *
Pheidole.7
Paratrechina.2
Anonychromyrma.1
Mesostruma.1             **
Iridomyrmex.5            ***
Iridomyrmex.2            *
Pristomyrmex.2
Monomorium.1
Papyrius.2               .
Camponotus.consobrinus
Ochetellus.1
Crematogaster.1
Mayriella.2              .
Tetramorium.3
Prolasius.6              .
Pheidole.7.1
Camponotus.12
Pachycondyla.2           *
Doleromyrma.1
Pheidole.6
Stigmacros.3             .
Myrmecia.3
Paratrechina.4
Heteroponera.1
Iridomyrmex.4
Technomyrmex.1
Polyrachis.8             ***
Prolasius.3
Dolichonderus.1
Machomyrma.3
Stigmacros.1
Stigmacros.4
Polyrachis.3
Rhopalomastix.1
Prolasius.1
Polyrachis.2
Myrmecia.2               *
```

```
Mayriella.1
Oligomyrmex.1             **
Iridomyrmex.6             **
Strumigenys.1
Solenopsis.1
Pachychondyla.1
Meranoplus.2              **
Papyrius.1
Prolasius.2
Melophorus.2
Pheidole.6.1              **
Oligomyrmex.2
Iridomyrmex.8             **
Myrmecia.1
Myrmecia.4                **
Rhopalomastix.5           **
Tetramorium.4
Anonychromyrma.2
Anonychromyrma.3
Camponotis.26
Camponotus.1
Colobostruma.1
Colobostruma.2
Crematogaster.3
Disturbed.lost
Doleromyrma.2
Epopostruma.1
Froggattella.1
Iridomyrmex.7
Leptomyrmex.1
Machomyrma.4
Machomyrma.6
Melophorus.1
Myrmecorhrynchus.1
Notoncus.3
Notoncus.4
Paratrechina.5
Paratrechina.6
Pheidole.1
Pheidole.2
Plagiolepis.1
Polyrachis.1
Polyrachis.5
Polyrachis.7
Pristomyrmex.1
Probolomyrmex.1
Prolasius.4
Prolasius.7
Rhopalomastix.2
Rhopalomastix.3
Stigmacros.2
Strumigenys.2
Tetramorium.5
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: CPW_SSTF

                           average       sd    ratio      ava      avb cumsum
Tapinoma.1                0.030263 0.028420 1.065000 0.733300 0.384600  0.044
Pheidole.5                0.029517 0.030650 0.963000 0.921900 1.081400  0.086
Meranoplus.1              0.029385 0.027290 1.076900 0.656600 0.153800  0.128
Iridomyrmex.purpureus     0.027624 0.031640 0.873200 0.542800 0.200300  0.168
Crematogaster.1           0.026790 0.027370 0.979000 0.250000 0.569600  0.207
Paratrechina.1            0.026602 0.027150 0.979800 0.578100 0.153800  0.245
```

Machomyrma.1	0.024562	0.025890	0.948600	0.437500	0.413700	0.281
Notoncus.1	0.023488	0.027970	0.839900	0.420600	0.245300	0.315
Anonychromyrma.1	0.022670	0.028920	0.783900	0.082300	0.494200	0.347
Pheidole.7	0.020393	0.025500	0.799600	0.344100	0.245300	0.377
Paratrechina.2	0.019403	0.024620	0.788000	0.250000	0.322200	0.405
Mesostruma.1	0.019143	0.026980	0.709500	0.394400	0.091500	0.432
Camponotus.consobrinus	0.018233	0.024860	0.733500	0.250000	0.245300	0.458
Ochetellus.1	0.018070	0.024570	0.735500	0.324300	0.168400	0.485
Pheidole.6	0.017694	0.023450	0.754400	0.187500	0.322200	0.510
Papyrius.2	0.016933	0.028240	0.599700	0.211200	0.153800	0.534
Prolasius.6	0.016813	0.024780	0.678600	0.315400	0.168400	0.559
Rhytidiponera..metallica.	0.016750	0.022810	0.734300	1.458600	1.226400	0.583
Iridomyrmex.5	0.016495	0.026310	0.627000	0.348000	0.000000	0.607
Doleromyrma.1	0.016024	0.023380	0.685200	0.199300	0.230800	0.630
Camponotus.12	0.014520	0.021640	0.670900	0.250000	0.153800	0.651
Pristomyrmex.2	0.014348	0.024190	0.593200	0.225200	0.153800	0.671
Mayriella.2	0.012995	0.021210	0.612800	0.250000	0.076900	0.690
Iridomyrmex.2	0.012878	0.023190	0.555400	0.273700	0.000000	0.709
Tetramorium.3	0.012610	0.025380	0.496800	0.000000	0.255100	0.727
Monomorium.1	0.011954	0.021280	0.561800	0.167600	0.200300	0.744
Stigmacros.3	0.010737	0.019980	0.537400	0.219100	0.076900	0.760
Pheidole.7.1	0.009809	0.021290	0.460700	0.199300	0.000000	0.774
Machomyrma.4	0.008692	0.021210	0.409900	0.000000	0.153800	0.786
Polyrachis.3	0.007763	0.016830	0.461200	0.125000	0.076900	0.797
Iridomyrmex.4	0.007139	0.019360	0.368800	0.093500	0.076900	0.808
Polyrachis.8	0.006228	0.016740	0.372100	0.125000	0.000000	0.817
Paratrechina.4	0.005458	0.014550	0.375000	0.062500	0.091500	0.825
Pachycondyla.2	0.005335	0.014540	0.366800	0.125000	0.000000	0.832
Doleromyrma.3	0.005103	0.013700	0.372500	0.125000	0.000000	0.840
Crematogaster.2	0.004783	0.017160	0.278700	0.000000	0.076900	0.847
Polyrachis.2	0.004583	0.012250	0.374000	0.125000	0.000000	0.853
Stigmacros.2	0.004238	0.015000	0.282500	0.000000	0.091500	0.859
Prolasius.7	0.004208	0.014990	0.280600	0.000000	0.076900	0.865
Anonychromyrma.2	0.003978	0.014140	0.281300	0.000000	0.076900	0.871
Polyrachis.7	0.003780	0.013410	0.281900	0.000000	0.076900	0.876
Stigmacros.1	0.003780	0.013410	0.281900	0.000000	0.076900	0.882
Pheidole.1	0.003758	0.013330	0.282000	0.000000	0.076900	0.887
Pristomyrmex.1	0.003564	0.012610	0.282500	0.000000	0.076900	0.892
Myrmecia.2	0.003360	0.013230	0.254000	0.062500	0.000000	0.897
Camponotis.26	0.003262	0.011510	0.283400	0.000000	0.076900	0.902
Camponotus.1	0.003262	0.011510	0.283400	0.000000	0.076900	0.907
Epopostruma.1	0.003262	0.011510	0.283400	0.000000	0.076900	0.911
Mayriella.1	0.003123	0.012280	0.254400	0.062500	0.000000	0.916
Oligomyrmex.1	0.003123	0.012280	0.254400	0.062500	0.000000	0.920
Anonychromyrma.3	0.003118	0.010990	0.283700	0.000000	0.076900	0.925
Myrmecorhrynchus.1	0.003118	0.010990	0.283700	0.000000	0.076900	0.929
Paratrechina.5	0.003118	0.010990	0.283700	0.000000	0.076900	0.934
Prolasius.3	0.003118	0.010990	0.283700	0.000000	0.076900	0.938
Oligomyrmex.2	0.003117	0.010990	0.283700	0.000000	0.076900	0.943
Prolasius.4	0.003117	0.010990	0.283700	0.000000	0.076900	0.947
Stigmacros.4	0.002945	0.011560	0.254800	0.062500	0.000000	0.952
Iridomyrmex.6	0.002910	0.011420	0.254800	0.062500	0.000000	0.956
Meranoplus.2	0.002732	0.010710	0.255100	0.062500	0.000000	0.960
Myrmecia.3	0.002732	0.010710	0.255100	0.062500	0.000000	0.964
Papyrius.1	0.002732	0.010710	0.255100	0.062500	0.000000	0.968
Heteroponera.1	0.002634	0.010290	0.255900	0.074300	0.000000	0.972
Rhopalomastix.1	0.002634	0.010290	0.255900	0.074300	0.000000	0.975
Melophorus.2	0.002413	0.009420	0.256200	0.074300	0.000000	0.979
Pheidole.6.1	0.002371	0.009270	0.255700	0.062500	0.000000	0.982
Prolasius.1	0.002215	0.008660	0.255900	0.062500	0.000000	0.985
Iridomyrmex.8	0.002029	0.007920	0.256200	0.062500	0.000000	0.988
Myrmecia.1	0.002029	0.007920	0.256200	0.062500	0.000000	0.991
Myrmecia.4	0.002029	0.007920	0.256200	0.062500	0.000000	0.994
Rhopalomastix.5	0.002029	0.007920	0.256200	0.062500	0.000000	0.997
Tetramorium.4	0.002029	0.007920	0.256200	0.062500	0.000000	1.000

```
Colobostruma.1            0.000000 0.000000      NaN 0.000000 0.000000  1.000
Colobostruma.2            0.000000 0.000000      NaN 0.000000 0.000000  1.000
Crematogaster.3           0.000000 0.000000      NaN 0.000000 0.000000  1.000
Disturbed.lost            0.000000 0.000000      NaN 0.000000 0.000000  1.000
Doleromyrma.2             0.000000 0.000000      NaN 0.000000 0.000000  1.000
Dolichonderus.1           0.000000 0.000000      NaN 0.000000 0.000000  1.000
Froggattella.1            0.000000 0.000000      NaN 0.000000 0.000000  1.000
Iridomyrmex.7             0.000000 0.000000      NaN 0.000000 0.000000  1.000
Leptomyrmex.1             0.000000 0.000000      NaN 0.000000 0.000000  1.000
Machomyrma.3              0.000000 0.000000      NaN 0.000000 0.000000  1.000
Machomyrma.6              0.000000 0.000000      NaN 0.000000 0.000000  1.000
Melophorus.1              0.000000 0.000000      NaN 0.000000 0.000000  1.000
Notoncus.3                0.000000 0.000000      NaN 0.000000 0.000000  1.000
Notoncus.4                0.000000 0.000000      NaN 0.000000 0.000000  1.000
Pachychondyla.1           0.000000 0.000000      NaN 0.000000 0.000000  1.000
Paratrechina.6            0.000000 0.000000      NaN 0.000000 0.000000  1.000
Pheidole.2                0.000000 0.000000      NaN 0.000000 0.000000  1.000
Plagiolepis.1             0.000000 0.000000      NaN 0.000000 0.000000  1.000
Polyrachis.1              0.000000 0.000000      NaN 0.000000 0.000000  1.000
Polyrachis.5              0.000000 0.000000      NaN 0.000000 0.000000  1.000
Probolomyrmex.1           0.000000 0.000000      NaN 0.000000 0.000000  1.000
Prolasius.2               0.000000 0.000000      NaN 0.000000 0.000000  1.000
Rhopalomastix.2           0.000000 0.000000      NaN 0.000000 0.000000  1.000
Rhopalomastix.3           0.000000 0.000000      NaN 0.000000 0.000000  1.000
Solenopsis.1              0.000000 0.000000      NaN 0.000000 0.000000  1.000
Strumigenys.1             0.000000 0.000000      NaN 0.000000 0.000000  1.000
Strumigenys.2             0.000000 0.000000      NaN 0.000000 0.000000  1.000
Technomyrmex.1            0.000000 0.000000      NaN 0.000000 0.000000  1.000
Tetramorium.5             0.000000 0.000000      NaN 0.000000 0.000000  1.000
                              p
Tapinoma.1                0.636
Pheidole.5                0.967
Meranoplus.1              0.295
Iridomyrmex.purpureus     0.035 *
Crematogaster.1           0.020 *
Paratrechina.1            0.251
Machomyrma.1              0.143
Notoncus.1                0.815
Anonychromyrma.1          0.890
Pheidole.7                0.945
Paratrechina.2            0.549
Mesostruma.1              0.013 *
Camponotus.consobrinus    0.059 .
Ochetellus.1              0.056 .
Pheidole.6                0.088 .
Papyrius.2                0.067 .
Prolasius.6               0.021 *
Rhytidiponera..metallica. 0.995
Iridomyrmex.5             0.002 **
Doleromyrma.1             0.209
Camponotus.12             0.100 .
Pristomyrmex.2            0.507
Mayriella.2               0.167
Iridomyrmex.2             0.350
Tetramorium.3             0.970
Monomorium.1              0.968
Stigmacros.3              0.070 .
Pheidole.7.1              0.565
Machomyrma.4              0.074 .
Polyrachis.3              0.152
Iridomyrmex.4             0.406
Polyrachis.8              0.134
Paratrechina.4            0.946
Pachycondyla.2            0.497
Doleromyrma.3             0.929
```

```
Crematogaster.2          0.919
Polyrachis.2             0.444
Stigmacros.2             0.193
Prolasius.7              0.201
Anonychromyrma.2         0.179
Polyrachis.7             0.193
Stigmacros.1             0.536
Pheidole.1               0.187
Pristomyrmex.1           0.663
Myrmecia.2               0.295
Camponotis.26            0.179
Camponotus.1             0.179
Epopostruma.1            0.179
Mayriella.1              0.420
Oligomyrmex.1            0.271
Anonychromyrma.3         0.166
Myrmecorhrynchus.1       0.166
Paratrechina.5           0.166
Prolasius.3              0.771
Oligomyrmex.2            0.371
Prolasius.4              0.179
Stigmacros.4             0.539
Iridomyrmex.6            0.258
Meranoplus.2             0.299
Myrmecia.3               0.886
Papyrius.1               0.693
Heteroponera.1           0.969
Rhopalomastix.1          0.859
Melophorus.2             0.556
Pheidole.6.1             0.288
Prolasius.1              0.850
Iridomyrmex.8            0.272
Myrmecia.1               0.718
Myrmecia.4               0.272
Rhopalomastix.5          0.272
Tetramorium.4            0.899
Colobostruma.1              NA
Colobostruma.2              NA
Crematogaster.3             NA
Disturbed.lost              NA
Doleromyrma.2               NA
Dolichonderus.1             NA
Froggattella.1              NA
Iridomyrmex.7               NA
Leptomyrmex.1               NA
Machomyrma.3                NA
Machomyrma.6                NA
Melophorus.1                NA
Notoncus.3                  NA
Notoncus.4                  NA
Pachychondyla.1             NA
Paratrechina.6              NA
Pheidole.2                  NA
Plagiolepis.1               NA
Polyrachis.1                NA
Polyrachis.5                NA
Probolomyrmex.1             NA
Prolasius.2                 NA
Rhopalomastix.2             NA
Rhopalomastix.3             NA
Solenopsis.1                NA
Strumigenys.1               NA
Strumigenys.2               NA
Technomyrmex.1              NA
Tetramorium.5               NA
```

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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: SRW_SSTF

                          average      sd   ratio     ava     avb cumsum     p
Rhytidiponera..metallica. 0.04104 0.03832 1.07100 0.79590 1.22640  0.055 0.044
Pheidole.5                0.03769 0.03488 1.08070 0.75730 1.08140  0.105 0.530
Anonychromyrma.1          0.03414 0.03631 0.94040 0.35530 0.49420  0.151 0.175
Crematogaster.1           0.03197 0.03498 0.91390 0.06250 0.56960  0.194 0.003
Notoncus.1                0.03025 0.03397 0.89040 0.48480 0.24530  0.234 0.226
Crematogaster.2           0.02819 0.03616 0.77980 0.43640 0.07690  0.272 0.003
Paratrechina.2            0.02653 0.03156 0.84050 0.33230 0.32220  0.307 0.034
Tetramorium.3             0.02548 0.03594 0.70910 0.29710 0.25510  0.341 0.353
Tapinoma.1                0.02482 0.02975 0.83420 0.18750 0.38460  0.375 0.980
Pheidole.7                0.02373 0.03238 0.73280 0.26180 0.24530  0.406 0.827
Doleromyrma.3             0.02273 0.04161 0.54640 0.33670 0.00000  0.437 0.045
Machomyrma.1              0.02239 0.02844 0.78740 0.06250 0.41370  0.467 0.306
Paratrechina.1            0.02126 0.02826 0.75220 0.34410 0.15380  0.495 0.764
Monomorium.1              0.02010 0.03262 0.61610 0.27240 0.20030  0.522 0.736
Meranoplus.1              0.01999 0.03189 0.62690 0.24430 0.15380  0.549 0.940
Pheidole.6                0.01891 0.02798 0.67590 0.06250 0.32220  0.574 0.041
Pristomyrmex.2            0.01695 0.02842 0.59640 0.21340 0.15380  0.597 0.312
Doleromyrma.1             0.01634 0.02805 0.58260 0.06250 0.23080  0.619 0.191
Iridomyrmex.purpureus     0.01478 0.03071 0.48120 0.06250 0.20030  0.638 0.783
Camponotus.consobrinus    0.01432 0.02443 0.58630 0.06250 0.24530  0.657 0.315
Ochetellus.1              0.01295 0.02735 0.47330 0.06250 0.16840  0.675 0.387
Papyrius.2                0.01185 0.02480 0.47770 0.06250 0.15380  0.691 0.369
Paratrechina.4            0.01184 0.02555 0.46330 0.12500 0.09150  0.707 0.569
Machomyrma.4              0.01147 0.02801 0.40940 0.00000 0.15380  0.722 0.002
Stigmacros.1              0.01049 0.02273 0.46140 0.12500 0.07690  0.736 0.023
Prolasius.3               0.01023 0.02214 0.46200 0.13680 0.07690  0.750 0.089
Iridomyrmex.2             0.00886 0.02459 0.36050 0.12500 0.00000  0.761 0.672
Camponotus.12             0.00855 0.02058 0.41560 0.00000 0.15380  0.773 0.654
Prolasius.6               0.00839 0.02076 0.40400 0.00000 0.16840  0.784 0.526
Technomyrmex.1            0.00803 0.02179 0.36870 0.12500 0.00000  0.795 0.092
Myrmecia.3                0.00801 0.02194 0.36520 0.12500 0.00000  0.805 0.380
Pachycondyla.2            0.00789 0.02131 0.37050 0.13680 0.00000  0.816 0.233
Heteroponera.1            0.00707 0.01918 0.36850 0.12500 0.00000  0.826 0.776
Dolichonderus.1           0.00704 0.01901 0.37050 0.12500 0.00000  0.835 0.105
Machomyrma.3              0.00691 0.01861 0.37110 0.12500 0.00000  0.844 0.527
Mayriella.2               0.00680 0.01792 0.37930 0.06250 0.07690  0.853 0.739
Iridomyrmex.4             0.00676 0.01782 0.37960 0.06250 0.07690  0.862 0.419
Oligomyrmex.2             0.00618 0.01621 0.38110 0.06250 0.07690  0.871 0.023
Mesostruma.1              0.00606 0.02144 0.28260 0.00000 0.09150  0.879 0.815
Prolasius.7               0.00548 0.01943 0.28180 0.00000 0.07690  0.886 0.001
Stigmacros.2              0.00528 0.01860 0.28390 0.00000 0.09150  0.893 0.002
Anonychromyrma.2          0.00510 0.01803 0.28260 0.00000 0.07690  0.900 0.001
Polyrachis.7              0.00478 0.01687 0.28330 0.00000 0.07690  0.906 0.002
Pheidole.1                0.00474 0.01674 0.28340 0.00000 0.07690  0.913 0.001
Pristomyrmex.1            0.00444 0.01564 0.28390 0.00000 0.07690  0.918 0.569
Camponotis.26             0.00398 0.01399 0.28470 0.00000 0.07690  0.924 0.003
Camponotus.1              0.00398 0.01399 0.28470 0.00000 0.07690  0.929 0.003
Epopostruma.1             0.00398 0.01399 0.28470 0.00000 0.07690  0.934 0.003
Polyrachis.3              0.00398 0.01399 0.28470 0.00000 0.07690  0.940 0.479
Anonychromyrma.3          0.00377 0.01324 0.28510 0.00000 0.07690  0.945 0.002
Myrmecorhrynchus.1        0.00377 0.01324 0.28510 0.00000 0.07690  0.950 0.002
Paratrechina.5            0.00377 0.01324 0.28510 0.00000 0.07690  0.955 0.002
Prolasius.4               0.00377 0.01323 0.28510 0.00000 0.07690  0.960 0.004
Prolasius.1               0.00367 0.01449 0.25330 0.06250 0.00000  0.965 0.634
Strumigenys.1             0.00344 0.01357 0.25380 0.06250 0.00000  0.970 0.501
Solenopsis.1              0.00342 0.01348 0.25380 0.06250 0.00000  0.974 0.774
Pachychondyla.1           0.00337 0.01328 0.25400 0.06250 0.00000  0.979 0.284
Rhopalomastix.1           0.00334 0.01313 0.25400 0.06250 0.00000  0.983 0.756
Pheidole.7.1              0.00318 0.01252 0.25430 0.06250 0.00000  0.987 0.956
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Prolasius.2              0.00318 0.01252 0.25430 0.06250 0.00000  0.992 0.543
Stigmacros.4             0.00318 0.01252 0.25430 0.06250 0.00000  0.996 0.411
Stigmacros.3             0.00311 0.01086 0.28600 0.00000 0.07690  1.000 0.728
Colobostruma.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Colobostruma.2           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Crematogaster.3          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Disturbed.lost           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Doleromyrma.2            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Froggattella.1           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.5            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.6            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.7            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Iridomyrmex.8            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Leptomyrmex.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Machomyrma.6             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Mayriella.1              0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Melophorus.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Melophorus.2             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Meranoplus.2             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.1               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.2               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Myrmecia.4               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Notoncus.3               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Notoncus.4               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Oligomyrmex.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Papyrius.1               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Paratrechina.6           0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.2               0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Pheidole.6.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Plagiolepis.1            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.1             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.2             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.5             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Polyrachis.8             0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Probolomyrmex.1          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.2          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.3          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Rhopalomastix.5          0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Strumigenys.2            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Tetramorium.4            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA
Tetramorium.5            0.00000 0.00000     NaN 0.00000 0.00000  1.000    NA

Rhytidiponera..metallica. *
Pheidole.5
Anonychromyrma.1
Crematogaster.1           **
Notoncus.1
Crematogaster.2           **
Paratrechina.2            *
Tetramorium.3
Tapinoma.1
Pheidole.7
Doleromyrma.3             *
Machomyrma.1
Paratrechina.1
Monomorium.1
Meranoplus.1
Pheidole.6                *
Pristomyrmex.2
Doleromyrma.1
Iridomyrmex.purpureus
Camponotus.consobrinus
Ochetellus.1
Papyrius.2
Paratrechina.4
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Machomyrma.4              **
Stigmacros.1              *
Prolasius.3               .
Iridomyrmex.2
Camponotus.12
Prolasius.6
Technomyrmex.1            .
Myrmecia.3
Pachycondyla.2
Heteroponera.1
Dolichonderus.1
Machomyrma.3
Mayriella.2
Iridomyrmex.4
Oligomyrmex.2             *
Mesostruma.1
Prolasius.7               ***
Stigmacros.2              **
Anonychromyrma.2          ***
Polyrachis.7              **
Pheidole.1                ***
Pristomyrmex.1
Camponotis.26             **
Camponotus.1              **
Epopostruma.1             **
Polyrachis.3
Anonychromyrma.3          **
Myrmecorhrynchus.1        **
Paratrechina.5            **
Prolasius.4               **
Prolasius.1
Strumigenys.1
Solenopsis.1
Pachychondyla.1
Rhopalomastix.1
Pheidole.7.1
Prolasius.2
Stigmacros.4
Stigmacros.3
Colobostruma.1
Colobostruma.2
Crematogaster.3
Disturbed.lost
Doleromyrma.2
Froggattella.1
Iridomyrmex.5
Iridomyrmex.6
Iridomyrmex.7
Iridomyrmex.8
Leptomyrmex.1
Machomyrma.6
Mayriella.1
Melophorus.1
Melophorus.2
Meranoplus.2
Myrmecia.1
Myrmecia.2
Myrmecia.4
Notoncus.3
Notoncus.4
Oligomyrmex.1
Papyrius.1
Paratrechina.6
Pheidole.2
Pheidole.6.1
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Plagiolepis.1
Polyrachis.1
Polyrachis.2
Polyrachis.5
Polyrachis.8
Probolomyrmex.1
Rhopalomastix.2
Rhopalomastix.3
Rhopalomastix.5
Strumigenys.2
Tetramorium.4
Tetramorium.5
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Permutation: free
Number of permutations: 999
```

Exercise: Which species of ants are the most important for the separation of communities?